Master table for the closing strategy: every 16:00-bar forecast, one scorer, the same days

Written by `experiments/master_table_close.py -arm-root linear_subsection_dedup` on 2026-10-18 17:26; every number below is read from `master_table.csv` (and `before_after.csv`, `recal_vs_plain.csv`) of the same run. Full table: `writeup/master_table_close.pdf`; tex: `writeup/generated/table_master_close.tex`. Every forecast is the plain back-transform $f^2 \cdot B$; every number is a point estimate.

Provenance: design, window mask, CPU class

• **Arm root:** `results/linear_subsection_dedup/` (de-duplicated design (16:00 campaign)); it supplies the arm-only rows, the per-bar OLS incumbent, the arm files the per-bar linear and VIX-only tables are gated against, and the common 16:00 target; its 16:00 target equals the default root's bit for bit (1469 rows, 0 differ; gate passed).

family	forecasts (A / B)	design	window mask	refit	tuning	CPU class	vs the pre-campaign table
paper	9 / 0	pooled 48-bar models (session-edge interactions kept, campaign decision 1)	n/a	as the paper	as the paper	not re-run	nothing
per-bar linear	22 / 0	de-duplicated per-bar design (commit 47f7f9c: the 12 HAR x open/close session-edge columns dropped)	identifiability mask at every penalty tune (unchanged: it had already removed the 12 columns)	every session	penalty every 250 sessions	Hoffman2 (agent A)	design de-dup (the mask had already dropped the session-edge columns: float path only)
pooled twin	21 / 0	pooled 48-bar twins (session-edge interactions kept)	identifiability mask	every session	penalty every 250 sessions	not re-run	nothing
VIX-only family	21 / 0	de-duplicated per-bar design (commit 47f7f9c: the 12 HAR x open/close session-edge columns dropped)	identifiability mask at every penalty tune (unchanged: it had already removed the 12 columns)	every session	penalty every 250 sessions	Hoffman2 (agent A)	design de-dup (the mask had already dropped the session-edge columns: float path only)
per-bar tree (untuned)	15 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every 10 sessions (T10)	none (shipped configuration)	CARC, pinned to one CPU class (epyc-7513)	design de-dup + per-window mask + CPU-class pinning (refit cadence unchanged, every 10)
per-bar tree (untuned, daily refit)	15 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session (T1)	none (shipped configuration)	CARC, pinned to one CPU class (epyc-7513)	new
per-bar tree (tuned)	24 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every 10 sessions (RS10)	random search, 32 candidates, every 250 sessions (MSE rule; tunedq = QLIKE rule)	CARC, pinned to one CPU class (epyc-7513)	design de-dup + per-window mask + CPU-class pinning (refit / tuning cadence unchanged)
per-bar tree (random search, daily)	24 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session (RS1)	random search, 32 candidates, every 250 sessions (MSE rule; tunedq = QLIKE rule)	CARC, pinned to one CPU class (epyc-7513)	new
per-bar tree (Optuna, TUNE_PER=1, best-of-50)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 50, every 1 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
per-bar tree (Optuna, TUNE_PER=5, best-of-50)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 50, every 5 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new

per-bar tree (Optuna, TUNE_PER=25, best-of-50)	24 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 50, every 25 session(s) (MSE rule; optunaq = QLIKE rule)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
per-bar tree (Optuna, TUNE_PER=250, best-of-50)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 50, every 250 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
per-bar tree (Optuna, TUNE_PER=1, best-of-25)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 25, every 1 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
per-bar tree (Optuna, TUNE_PER=25, best-of-25)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 25, every 25 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
per-bar tree (Optuna, TUNE_PER=1, best-of-10)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 10, every 1 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
per-bar tree (Optuna, TUNE_PER=25, best-of-10)	12 / 0	de-duplicated per-bar design	per-window mask (src/models/window_mask.py, WINDOW_MASK=1)	every session	Optuna TPE, 50 trials, best of the first 10, every 25 session(s)	CARC, single-class re-run on epyc-7513 (agent C; canonical)	new
direct rest-of-day at 15:30 (check)	9 / 0	de-duplicated per-bar design, 15:30 clock of the rest-of-day model	identifiability mask	every session	penalty every 250 sessions	Hoffman2, each 15:30 arm pinned to its per-bar twin's CPU architecture (agent E)	design de-dup + CPU-architecture pinning to the per-bar twin
implied-vol representations	15 / 0	de-duplicated per-bar design (commit 47f7f9c: the 12 HAR x open/close session-edge columns dropped)	identifiability mask at every penalty tune (unchanged: it had already removed the 12 columns)	every session	penalty every 250 sessions	Hoffman2 (agent A)	design de-dup (the mask had already dropped the session-edge columns: float path only)
HAR-ladder variants	9 / 0	de-duplicated per-bar design (commit 47f7f9c: the 12 HAR x open/close session-edge columns dropped)	identifiability mask at every penalty tune (unchanged: it had already removed the 12 columns)	every session	penalty every 250 sessions	Hoffman2 (agent A)	design de-dup (the mask had already dropped the session-edge columns: float path only)
chain-period buckets	0 / 21	de-duplicated per-bar design (commit 47f7f9c: the 12 HAR x open/close session-edge columns dropped)	identifiability mask at every penalty tune (unchanged: it had already removed the 12 columns)	every session	penalty every 250 sessions	Hoffman2 (agent A)	design de-dup (the mask had already dropped the session-edge columns: float path only)

Scorer, days, reference

• **Scorer: the plain back-transform.** Every forecast is used as $f^2 \cdot B$ (f = the model's forecast on the adjusted scale sqrt(RV/B), B = the diurnal baseline; the `yhat`^{2} · `baseline` of `results/spxw_pnl/yhat_*.parquet`); QLIKE (`qlike_raw`) against the ACTUAL 15:30-16:00 realized variance (`rv_unclipped`; since 2026-10-15, data audit #10 – the models still fit the spec's winsorized target, whose QLIKE is the diagnostic `qlike_raw_trimmed`); the trade is the deck's 15:30 sign(s) on the straddle (`trade_1530`): buy when $f^2 \cdot B$ exceeds the 15:30 implied variance, sell otherwise. The research scorer's 16:00-bar recalibration $(f^2 + s) \cdot B$ (s = the forecast's own trailing 250-session mean squared adjusted-scale error, lagged one) was removed on 2026-10-06 (user decision); it survives only as the diagnostic column `pred_clock` and in `recal_vs_plain.csv` (section below). The notebook's 13-bar Mincer–Zarnowitz map is **not** used either: no number here may be set beside a notebook number.

• **Point estimates only.** The circular block bootstrap is disabled (commit b761b28: every draw is the original sample), so no interval is reported. The inference shown is the Diebold–Mariano statistic on the daily QLIKE difference (`dm_test`, Newey–West variance with the Harvey–Leybourne–Newbold correction) and the Newey–West HAC t of the daily mid-return difference (`newey_west_t`); counts 'significant' use the two-sided 5 % level of those statistics.

• **Days:** table A = 295 forecasts in 20 families (+ 9 check rows) on the same 866 trade days (2020-01-03 .. 2024-04-30); intersection over table A = 866 of 866 trade days. Every family that covers the 866 trade days is in table A. Table B = 21 forecasts scored on their own days (paired with the reference on those days): 21

chain-period buckets forecasts from 2022-01-03 on 550 trade days. Forecasts that failed to load: none.

• **Which forecasts drop days:** no table-A forecast misses a trade day; 0 forecasts miss trade days inside their own window. The 21 table-B forecasts start at their comparison window (2022-01-03), so each omits the 316 earlier trade days by design (their implied-vol inputs are real chain data only from then; compare_mfiv_harlag.CHAIN_START). Coverage per forecast: `master_table_days.csv`.

• **Reference:** the block-diagonal ridge (`blk2`, `yhat_blk2_fomc1.parquet`) — the paper's headline forecast (HAR ladder, penalty 1, plus the exogenous block, penalty 100, on the panel of record), the forecast the paper's 15:30 deck trades. Under this scorer it has QLIKE 0.1092 and sign(s) Sharpe 1.28 mid / 0.80 crossed (the deck's own map gives it 1.34 mid; the difference is the scorer). Second reference: always short, Sharpe 0.20 mid / -0.27 crossed.

(a) Recommended headline forecast

per-bar ridge [live_feasible] (`sub_ridge_live_feasible`), scored with the plain forecast. On the 866 days: QLIKE 0.1206 (+10.4 % vs the reference, DM +0.59, p 0.558); sign(s) Sharpe 1.79 mid / 1.31 crossed, mean +0.126 per unit premium (crossed +0.093), hit rate 57.0 %, buys on 269 days (31.1 %). Sharpe difference vs the reference: mid +0.51, crossed +0.51 (HAC t +1.12, p 0.262 on the daily mid-return difference); vs always short: mid +1.59, crossed +1.58 (HAC t +2.26, p 0.024).

• **Does any forecast now beat the headline?** Of the 294 other table-A forecasts (check rows and exact duplicates left out), 30 have a higher Sharpe point estimate at the mid fill (31 crossed); with the HAC t of the daily mid-return difference significant at 5 %: 2 (per-bar ridge [live_vix_only] (+0.73, HAC t +2.46, p 0.014); per-bar ridge [free_vix_only] (+0.70, HAC t +2.45, p 0.014)). 282 have a lower QLIKE point estimate; with DM $p < 0.05$: 0.

Highest Sharpe differences over the headline (mid, point estimates): per-bar ridge [live_vix_only] +0.73 (t +2.46, p 0.014); per-bar ridge [free_vix_only] +0.70 (t +2.45, p 0.014); Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [live_feasible] +0.40 (t +1.05, p 0.292); per-bar ridge [lean_vol_vix] +0.31 (t +0.83, p 0.407); per-bar ridge [free_feasible] +0.28 (t +1.14, p 0.253).

RECORDED, not acted on: the headline constant stays `sub_ridge_live_feasible` (it is chosen on feasibility and as the per-bar model of record, not on the maximum of many Sharpe ratios); the forecasts above are the candidates to look at.

Why this one:

• It is a 15:30-specific model on the inputs a 15:30 forecaster can rebuild live (`live_feasible`), so the trade it scores can be run. It ranks 31 of 295 table-A forecasts on sign(s) Sharpe (mid) and 283 of 295 on QLIKE (plain forecast).

• The highest Sharpe in table A is per-bar ridge [live_vix_only] (2.53 mid / 2.05 crossed). Choosing the maximum of 295 Sharpe ratios would select on the trade's own noise; the headline is chosen on feasibility and on being the per-bar model of record (the per-bar ridge is the estimator the 15:30 study defined first), not on the maximum.

• Paired against the headline itself on the same days (`master_table_vs_headline.csv`; a negative QLIKE % / DM = the alternative forecasts better, a positive Sharpe difference = the alternative trades better; point estimates, HAC t of the daily mid-return difference):

alternative	QLIKE	% vs headline	DM	Sharpe mid / crossed	ΔSharpe mid vs headline	ΔSharpe crossed vs headline	HAC t (mid)	same position
per-bar lasso [live_feasible]	0.0950	-21.2	-1.30	1.87 / 1.39	+0.08	+0.08	+0.19	90.2 %
per-bar elastic net [live_feasible]	0.0973	-19.3	-1.25	1.56 / 1.08	-0.23	-0.22	-0.55	91.7 %
per-bar ridge [all_features]	0.1184	-1.8	-0.26	1.71 / 1.22	-0.08	-0.08	-0.18	87.4 %
per-bar ridge [baseline]	0.0968	-19.7	-1.16	1.36 / 0.88	-0.43	-0.43	-0.77	83.9 %
per-bar lasso [free_vix_only]	0.0959	-20.5	-1.25	1.91 / 1.43	+0.12	+0.12	+0.29	89.8 %
per-bar ridge [free_vix_only]	0.1216	+0.8	+0.26	2.49 / 2.02	+0.70	+0.71	+2.45	96.5 %
per-bar LightGBM [live_feasible]	0.0959	-20.5	-1.27	1.54 / 1.05	-0.25	-0.25	-0.60	86.3 %
per-bar XGBoost [live_feasible]	0.0966	-19.9	-1.23	1.78 / 1.29	-0.01	-0.01	-0.03	87.5 %
per-bar XGBoost [lean_vol_vix]	0.0943	-21.8	-1.37	1.78 / 1.30	-0.01	-0.01	-0.02	87.5 %
per-bar XGBoost, daily refit [lean_vol_vix]	0.0932	-22.7	-1.40	1.76 / 1.28	-0.03	-0.03	-0.07	87.2 %
tuned per-bar XGBoost [live_feasible], MSE-selected	0.1025	-15.0	-1.00	1.96 / 1.48	+0.17	+0.17	+0.36	85.3 %
tuned per-bar LightGBM [live_feasible], MSE-selected	0.0954	-20.9	-1.28	1.76 / 1.28	-0.03	-0.03	-0.06	86.8 %
tuned per-bar XGBoost, daily refit [live_feasible], QLIKE-selected	0.1008	-16.4	-1.03	1.99 / 1.51	+0.20	+0.21	+0.43	85.8 %

tuned per-bar random forest, daily refit [lean_vol_vix], QLIKE-selected	0.0949	-21.3	-1.27	1.47 / 0.98	-0.32	-0.33	-0.59	88.9 %
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [live_feasible]	0.0995	-17.5	-1.05	2.19 / 1.70	+0.40	+0.39	+1.05	85.7 %
Optuna per-bar random forest, TUNE_PER 250, best of 50 [lean_vol_vix]	0.0953	-20.9	-1.25	1.29 / 0.80	-0.50	-0.51	-1.01	89.0 %

• Over all 294 other table-A forecasts (point estimates): 30 trade better than the headline at the mid fill (31 crossed), 2 of them with the HAC t of the daily mid-return difference significant at 5 %; 264 trade worse (30 significant); 282 have a lower QLIKE (0 with DM $p < 0.05$), 12 a higher one (3 with DM $p < 0.05$).

• Against the reference, 148 of 294 table-A forecasts have a higher mid-fill Sharpe (147 at the crossed fill); 4 of them with the HAC t of the daily mid-return difference significant at 5 %. Against always short, 293 (mid) and 293 (crossed) of 295 have a higher Sharpe; 109 with a significant HAC t (mid).

• On forecast accuracy alone the best live-feasible forecast is per-bar LightGBM, daily refit [live_feasible] (QLIKE 0.0933, Sharpe 1.61 / 1.12); a QLIKE-first choice would take it. The trade does not rank forecasts the way QLIKE does (section b), which is why the headline names the scorer AND the trade numbers.

Scorer change: recalibrated -> plain (2026-10-06)

The recalibration was removed on 2026-10-06 (user decision): every number in this file uses the plain forecast $f^2 \cdot B$. `recal_vs_plain.csv` records, per forecast on its own scored days, the parked recalibrated forecast (`pred_clock` = $(f^2 + s) \cdot B$, diagnostic) against the plain one. Over the 286 table-A forecasts (check rows and exact duplicates left out, 866 days):

• QLIKE, plain minus recalibrated: median -4.8 (-8.0 .. +38.4) %; Sharpe mid median -0.04 (-0.99 .. +0.71), crossed median -0.06 (-1.01 .. +0.70); buy share median -9.6 (-12.5 .. -6.9) percentage points.

• Positions changed: median 83 (60 .. 108) of 866 days; buy -> sell 23527, sell -> buy 0 over all of them ($s \cdot B \geq 0$, so removing it can only lower a forecast: mean $s \cdot B$ / $f^2 \cdot B$ median 0.08 (0.08 .. 0.18)).

• **Headline** (`sub_ridge_live_feasible`): QLIKE 0.1005 -> 0.1184; Sharpe mid 1.90 -> 1.79, crossed 1.44 -> 1.31; buys 40.2 % -> 31.1 %; positions changed on 79 days.

• **Reference** (`blk2`): QLIKE 0.1058 -> 0.1068; Sharpe mid 1.00 -> 1.28, crossed 0.53 -> 0.80; buys 36.0 % -> 27.0 %; positions changed on 78 days.

• Largest Sharpe (mid) falls: Optuna per-bar random forest, TUNE_PER 1, best of 25 [all_features] 1.56 -> 0.57; per-bar lasso [vix_rvol] 1.63 -> 0.79; per-bar XGBoost, daily refit [all_features] 1.87 -> 1.12. Largest rises: per-bar elastic net [all_features] 1.33 -> 2.04; per-bar ridge [free_vix_only] 1.83 -> 2.49; per-bar ridge [live_vix_only] 1.86 -> 2.53.

• Rank agreement across these forecasts, recalibrated vs plain: Spearman +0.80 on QLIKE, +0.67 on Sharpe mid.

Before / after the 16:00 campaign (both sides plain)

Before = the pre-campaign master table (commit 6177b74; snapshot `results/spxw_pnl/pre_dedup_2026-09-29/`), re-scored here with the plain forecast from its daily frame's `plain` column; after = this run; every key present in both is paired on the same days (`before_after.csv`; difference = after minus before; point estimates, HAC t of the daily mid-return difference).

• **Headline** (`sub_ridge_live_feasible`; design de-dup (the mask had already dropped the session-edge columns: float path only); re-run on hoffman2 (was carc)): QLIKE 0.1206 -> 0.1206 (+0.00 %); Sharpe mid 1.79 -> 1.79 (+0.00, HAC t n/a), crossed 1.31 -> 1.31 (+0.00); positions changed on 0 of 866 days; largest relative change of the plain forecast 1.13e-12.

• **Reference** (`blk2`; nothing): QLIKE 0.1092 -> 0.1092 (+0.00 %); Sharpe mid 1.28 -> 1.28 (+0.00, HAC t n/a), crossed 0.80 -> 0.80 (+0.00); positions changed on 0 of 866 days; largest relative change of the plain forecast 0.00e+00.

family	keys in both	what changed	QLIKE % change, median (min .. max)	ΔSharpe mid, median (min .. max)	ΔSharpe crossed, median (min .. max)	HAC t significant, up / down	positions changed, median (max)
paper	9	nothing; nothing (the campaign did not re-run it	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	0 / 0	0 (0)
per-bar linear	10	design de-dup (the mask had already dropped the session-edge columns: float path only)	+0.00 (-0.00 .. +0.04)	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	0 / 0	0 (0)
pooled twin	9	nothing	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	0 / 0	0 (0)
VIX-only family	21	design de-dup (the mask had already dropped the session-edge columns: float path only)	+0.00 (-0.03 .. +0.05)	+0.00 (-0.01 .. +0.00)	+0.00 (-0.01 .. +0.00)	0 / 0	0 (1)
per-bar tree (untuned)	9	design de-dup + per-window mask + CPU-class pinning (refit cadence unchanged, every 10)	+0.71 (-1.80 .. +1.58)	-0.10 (-0.62 .. +0.05)	-0.10 (-0.62 .. +0.05)	0 / 2	37 (61)
per-bar tree (tuned)	18	design de-dup + per-window mask + CPU-class pinning (refit / tuning cadence unchanged)	+1.59 (-5.24 .. +13.14)	+0.01 (-0.69 .. +0.48)	+0.01 (-0.69 .. +0.48)	0 / 2	60 (89)
direct rest-of-day at 15:30 (check)	9	design de-dup + CPU-architecture pinning to the per-bar twin	-0.00 (-0.00 .. +0.01)	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	0 / 0	0 (0)
implied-vol representations	15	design de-dup (the mask had already dropped the session-edge columns: float path only)	+0.00 (-0.00 .. +0.05)	+0.00 (+0.00 .. +0.02)	+0.00 (+0.00 .. +0.02)	0 / 0	0 (1)
HAR-ladder variants	9	design de-dup (the mask had already dropped the session-edge columns: float path only)	+0.00 (-0.01 .. +0.02)	+0.00 (-0.05 .. +0.00)	+0.00 (-0.05 .. +0.00)	0 / 0	0 (2)
chain-period buckets	21	design de-dup (the mask had already dropped the session-edge columns: float path only)	-0.00 (-0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	+0.00 (+0.00 .. +0.00)	0 / 0	0 (0)

Keys whose own mid-fill return moved with the HAC t significant at 5 % (Sharpe before -> after): tuned per-bar LightGBM [all_features], QLIKE-selected 2.23 -> 1.54 (t -2.45, p 0.014); per-bar random forest [all_features] 1.89 -> 1.27 (t -2.35, p 0.019); tuned per-bar LightGBM [baseline], MSE-selected 1.45 -> 0.95 (t -3.11, p 0.002); per-bar LightGBM [baseline] 1.19 -> 0.73 (t -2.28, p 0.023).

(b) QLIKE vs Sharpe across models

• **paper** (9): QLIKE 0.1078 .. 0.1270, Sharpe mid 0.43 .. 1.50; best QLIKE lasso (fixed 1e-4), earlier panel [on disk, unused by the paper] (0.1078, Sharpe 1.36); best Sharpe lasso (causally tuned) (1.50, QLIKE 0.1115).

• **per-bar linear** (22): QLIKE 0.0931 .. 0.1206, Sharpe mid 1.23 .. 2.11; best QLIKE per-bar lasso [all_features], pre-tuned (0.0931, Sharpe 1.88); best Sharpe per-bar ridge [lean_vol_vix] (2.11, QLIKE 0.0985).

• **pooled twin** (21): QLIKE 0.1019 .. 0.1098, Sharpe mid 0.69 .. 1.50; best QLIKE pooled elastic net [all_features], same spec (0.1019, Sharpe 1.15); best Sharpe

pooled elastic net [all_features], pre-tuned (1.50, QLIKE 0.1040).

• **VIX-only family** (21): QLIKE 0.0944 .. 0.1431, Sharpe mid 0.74 .. 2.53; best QLIKE per-bar lasso [vix_rvol] (0.0944, Sharpe 0.79); best Sharpe per-bar ridge [live_vix_only] (2.53, QLIKE 0.1201).

• **per-bar tree (untuned)** (15): QLIKE 0.0943 .. 0.1018, Sharpe mid 0.73 .. 1.78; best QLIKE per-bar XGBoost [lean_vol_vix] (0.0943, Sharpe 1.78); best Sharpe per-bar XGBoost [lean_vol_vix] (1.78, QLIKE 0.0943).

• **per-bar tree (untuned, daily refit)** (15): QLIKE 0.0932 .. 0.1010, Sharpe mid 0.66 .. 1.76; best QLIKE per-bar XGBoost, daily refit [lean_vol_vix] (0.0932, Sharpe 1.76); best Sharpe per-bar XGBoost, daily refit [lean_vol_vix] (1.76, QLIKE 0.0932).

• **per-bar tree (tuned)** (24): QLIKE 0.0954 .. 0.1141, Sharpe mid 0.65 .. 1.96; best QLIKE tuned per-bar LightGBM [live_feasible], MSE-selected (0.0954, Sharpe 1.76); best Sharpe tuned per-bar XGBoost [live_feasible], MSE-selected (1.96, QLIKE 0.1025).

• **per-bar tree (random search, daily)** (24): QLIKE 0.0949 .. 0.1120, Sharpe mid 0.46 .. 1.99; best QLIKE tuned per-bar random forest, daily refit [lean_vol_vix], QLIKE-selected (0.0949, Sharpe 1.47); best Sharpe tuned per-bar XGBoost, daily refit [live_feasible], QLIKE-selected (1.99, QLIKE 0.1008).

• **per-bar tree (Optuna, TUNE_PER=1, best-of-50)** (12): QLIKE 0.0966 .. 0.1055, Sharpe mid 0.59 .. 1.50; best QLIKE Optuna per-bar random forest, TUNE_PER 1, best of 50 [lean_vol_vix] (0.0966, Sharpe 1.50); best Sharpe Optuna per-bar random forest, TUNE_PER 1, best of 50 [lean_vol_vix] (1.50, QLIKE 0.0966).

• **per-bar tree (Optuna, TUNE_PER=5, best-of-50)** (12): QLIKE 0.0964 .. 0.1044, Sharpe mid 0.41 .. 1.48; best QLIKE Optuna per-bar random forest, TUNE_PER 5, best of 50 [lean_vol_vix] (0.0964, Sharpe 1.48); best Sharpe Optuna per-bar random forest, TUNE_PER 5, best of 50 [lean_vol_vix] (1.48, QLIKE 0.0964).

• **per-bar tree (Optuna, TUNE_PER=25, best-of-50)** (24): QLIKE 0.0958 .. 0.1064, Sharpe mid 0.45 .. 2.19; best QLIKE Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix] (0.0958, Sharpe 1.24); best Sharpe Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [live_feasible] (2.19, QLIKE 0.0995).

• **per-bar tree (Optuna, TUNE_PER=250, best-of-50)** (12): QLIKE 0.0953 .. 0.1100, Sharpe mid 0.64 .. 1.88; best QLIKE Optuna per-bar random forest, TUNE_PER 250, best of 50 [lean_vol_vix] (0.0953, Sharpe 1.29); best Sharpe Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [live_feasible] (1.88, QLIKE 0.1015).

• **per-bar tree (Optuna, TUNE_PER=1, best-of-25)** (12): QLIKE 0.0969 .. 0.1058, Sharpe mid 0.46 .. 1.43; best QLIKE Optuna per-bar random forest, TUNE_PER 1, best of 25 [lean_vol_vix] (0.0969, Sharpe 1.06); best Sharpe Optuna per-bar random forest, TUNE_PER 1, best of 25 [live_feasible] (1.43, QLIKE 0.0991).

• **per-bar tree (Optuna, TUNE_PER=25, best-of-25)** (12): QLIKE 0.0974 .. 0.1040, Sharpe mid 0.33 .. 1.31; best QLIKE Optuna per-bar random forest, TUNE_PER 25, best of 25 [lean_vol_vix] (0.0974, Sharpe 1.31); best Sharpe Optuna per-bar random forest, TUNE_PER 25, best of 25 [lean_vol_vix] (1.31, QLIKE 0.0974).

• **per-bar tree (Optuna, TUNE_PER=1, best-of-10)** (12): QLIKE 0.0975 .. 0.1051, Sharpe mid 0.59 .. 1.57; best QLIKE Optuna per-bar random forest, TUNE_PER 1, best of 10 [lean_vol_vix] (0.0975, Sharpe 1.10); best Sharpe Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [lean_vol_vix] (1.57, QLIKE 0.0996).

• **per-bar tree (Optuna, TUNE_PER=25, best-of-10)** (12): QLIKE 0.0962 .. 0.1076, Sharpe mid 0.29 .. 1.71; best QLIKE Optuna per-bar random forest, TUNE_PER 25, best of 10 [lean_vol_vix] (0.0962, Sharpe 1.41); best Sharpe Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [lean_vol_vix] (1.71, QLIKE 0.1029).

• **implied-vol representations** (15): QLIKE 0.0941 .. 0.1166, Sharpe mid 1.25 .. 1.90; best QLIKE per-bar elastic net [ivrep_vrp] (0.0941, Sharpe 1.59); best Sharpe per-bar elastic net [ivrep_term] (1.90, QLIKE 0.0966).

• **HAR-ladder variants** (9): QLIKE 0.0950 .. 0.2333, Sharpe mid -0.12 .. 2.02; best QLIKE per-bar lasso [live_feasible], HAR ladder base2 (0.0950, Sharpe 2.02); best Sharpe per-bar lasso [live_feasible], HAR ladder base2 (2.02, QLIKE 0.0950).

• Over all of table A the lowest QLIKE is per-bar lasso [all_features], pre-tuned (0.0931; Sharpe 1.88, rank 20 of 295 on Sharpe) and the highest Sharpe is per-bar

ridge [live_vix_only] (2.53; QLIKE 0.1201, rank 282 of 295 on QLIKE). Median over table A: QLIKE 0.1010, Sharpe mid 1.28, crossed 0.80.

• Against the reference's QLIKE, 254 forecasts have a lower point estimate (better), 135 of them with DM $p < 0.05$; 40 a higher one, 6 with DM $p < 0.05$.

Rank correlation across table-A forecasts (check rows and exact duplicates excluded), point estimate (a negative value = lower QLIKE goes with higher Sharpe):

set	forecasts	QLIKE	Sharpe	Spearman
all table-A forecasts	295	plain	mid	-0.29
all table-A forecasts	295	plain	crossed	-0.29
per-bar linear + VIX-only family	43	plain	mid	+0.45
per-bar linear + VIX-only family	43	plain	crossed	+0.45
48-bar forecasts (paper + pooled twins)	30	plain	mid	-0.42
48-bar forecasts (paper + pooled twins)	30	plain	crossed	-0.44

(c) Extreme back-transform values

• **Rule (named):** PLAIN_FLOOR(d) = the smallest 16:00 target of the previous 250 sessions (FLOOR_W, the parked recalibration's window), lagged one session. A plain forecast $f^2 \cdot B$ below it forecasts a quieter 15:30–16:00 half hour than any of the past year; it is flagged, listed in `master_table_extremes.csv`, and the QLIKE is reported as is (`qlike_raw`, the reported number) and, as a diagnostic, with flagged forecasts raised to the floor (`qlike_raw_floored`). Nothing is clipped.

• **On the trade days:** 0 flagged forecasts over all forecasts, so `qlike_raw_floored` = `qlike_raw` for 325 of 325 rows.

• **On every 16:00 session of the arm span** (1406 sessions, the arm span 2018-06-25 .. 2024-04-30 after the floor's 63-session warm-up, early closes included): 448 flagged rows over 177 forecasts, 448 of them on 8 early-close sessions (2019-07-03, 2019-11-29, 2019-12-24, 2020-11-27, 2020-12-24, 2022-11-25, 2023-07-03, 2023-11-24): 13:00 closes, where the 15:30–16:00 bar lies after the cash close and the trade frame never enters. Negative adjusted-scale forecasts occur there too (the plain back-transform squares them).

• per-bar ridge [all_features] on 2023-11-24 (early close): f = 0.0047 on the adjusted scale, plain forecast 5.51e-11 vs floor 2.14e-07 and realized variance 1.94e-07; that one day's QLIKE is 3516.8. Over all 1406 sessions the QLIKE is 2.6480 as is, 0.1392 floored, 0.1344 without early closes; on the trade days 0.1184.

• per-bar ridge [live_feasible_plus_ivslice], 500 sessions on 2023-07-03 (early close): f = -0.0185 on the adjusted scale, plain forecast 1.05e-09 vs floor 2.14e-07 and realized variance 4.56e-07; that one day's QLIKE is 427.2. Over all 1406 sessions the QLIKE is 0.4471 as is, 0.1391 floored, 0.1357 without early closes; on the trade days 0.0883.

• The handoff's '16:00 ridge QLIKE 2.65 unrecalibrated' is this same early-close day: `score_rest_of_day.csv` reports the plain QLIKE 2.6480 for the all_features ridge at the 15:30 clock on 1406 sessions against the unclipped variance; the trade frame excludes early closes, so it does not enter any number of this table.

Gates

1877 gates checked, 0 failed (`master_table_gates.csv`, column `asserted`). Among them: every arm file's 16:00 target equals the common target and the common target of `linear_subsection_dedup` equals `linear_subsection`'s bit for bit; every table's baseline equals its B, its rv_raw the production table's, and its 16:00 stamps the target's; every arm file's plain forecast equals the research reader's $f^2 \cdot B$ on the trade days; every per-bar linear and VIX-only table equals its arm file; every row's trade aggregates equal `trade_1530`'s on the plain forecast; every row of `master_table_daily.parquet` has q = sign(plain - iv_var) and the daily frame reproduces every row's Sharpe, QLIKE and buys; no table falls into 'other table'; the paper's always-short row is reproduced; the families the campaign did not touch equal the pre-campaign table under the plain scorer; stored plain numbers (16:00 campaign, plain forecast: 45; paper: 2; pre-campaign master table (plain): 17).

Reported, not asserted (2000): stored numbers written with the recalibrated forecast (compared with the same quantity recomputed here from the diagnostic `pred_clock`) or on another design — : 0 of 1 agree to 1e-09; 16:00 campaign, plain forecast: 0 of 36 agree to 1e-09; 16:00 campaign, recalibrated scorer: 530 of 770 agree to 1e-09; committed recalibrated master table (2026-09-30): 756 of 1000 agree to 1e-09; pre-campaign design: 166 of 192 agree to 1e-09; pre-campaign master table (plain): 1 of 1 agree to 1e-09. A difference there is a design change or an input that changed since the stored number was written, not a failure.

Top of table A by sign(s) Sharpe (mid)

forecast	QLIKE	Δ% vs ref	DM	Sharpe mid	Sharpe crossed	ΔSharpe mid vs ref	HAC t vs ref	ΔSharpe mid vs short	buy days
per-bar ridge [live_vix_only]	0.1201	+10.0	+0.55	2.53	2.05	+1.24	+2.78	+2.32	272
per-bar ridge [free_vix_only]	0.1216	+11.4	+0.57	2.49	2.02	+1.21	+2.65	+2.29	271
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [live_feasible]	0.0995	-8.9	-2.46	2.19	1.70	+0.91	+1.97	+1.99	231
per-bar ridge [lean_vol_vix]	0.0985	-9.8	-1.94	2.11	1.63	+0.83	+1.87	+1.90	257
per-bar ridge [free_feasible]	0.1431	+31.0	+0.90	2.08	1.59	+0.79	+1.47	+1.87	254
per-bar elastic net [all_features]	0.1006	-7.9	-2.10	2.04	1.56	+0.76	+1.63	+1.84	256
per-bar ridge [live_vix_rvol]	0.1156	+5.9	+0.40	2.04	1.56	+0.76	+1.66	+1.83	272
per-bar ridge [free_feasible_vol]	0.1220	+11.7	+0.60	2.03	1.55	+0.75	+1.51	+1.83	268
per-bar lasso [live_feasible], HAR ladder base2	0.0950	-13.0	-3.76	2.02	1.54	+0.74	+2.05	+1.82	237
per-bar lasso [all_features]	0.0996	-8.8	-2.80	2.00	1.53	+0.72	+1.53	+1.80	259
tuned per-bar XGBoost, daily refit [live_feasible], QLIKE-selected	0.1008	-7.7	-1.82	1.99	1.51	+0.71	+1.49	+1.79	252
tuned per-bar XGBoost [live_feasible], MSE-selected	0.1025	-6.2	-1.40	1.96	1.48	+0.68	+1.48	+1.75	260

Wording: sign(s) = buy the straddle when the plain 16:00 forecast $f^2 \cdot B$ exceeds the 15:30 implied variance, sell otherwise; straddle = nearest out-of-the-money call + nearest out-of-the-money put, same-day expiry, one position; the last-30-min trade enters at 15:30 and settles at the close. Buckets: `baseline` = HAR + calendar (`har_ma_*`, calendar dummies), `all_features` = the full design, `live_feasible` = the 16 series a 15:30 forecaster rebuilds live (ES return moments and liquidity, VIX/VVIX/VIX3M (`vix_ma_*` ...), FOMC calendar), `vix_only` = HAR + calendar + VIX level, `live_vix_only` = live_feasible minus VVIX and VIX3M, `free_vix_only` = the free feed (ES 1-min bars + VIX + FOMC calendar). Tree rungs: T10 / T1 = the shipped configuration refit every 10 sessions / every session; RS10 / RS1 = random search (32 candidates, every 250 sessions) refit every 10 / every session; Optuna = TPE, 50 trials per tuning point, tuning every TUNE_PER sessions, best of the first k trials, refit every session.

Table 6: Forecast accuracy at the 16:00 bar (15:30–16:00). Every forecast is the plain back-transform $f^2 B$ (f the forecast on the adjusted scale $\sqrt{RV/B}$, B the diurnal baseline). QLIKE: against the actual (unwinsorized) realized variance; QLIKE, trimmed: the same forecast against the winsorized training target (diagnostic). % vs R: the mean daily loss difference against the reference R; DM: Diebold–Mariano statistic on that difference, Newey–West variance (Bartlett, lag 6) with the Harvey–Leybourne–Newbold correction; $\overline{RV}/\overline{F}$: mean realized variance over mean forecast; extreme: plain forecasts below PLAIN_FLOOR on every 16:00 session of the arm span. Point estimates. R = reference, H = recommended headline, = identical to another row.

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
Table A: the same 866 trade days, 2020-01-03 to 2024-04-30						
paper						
baseline (HAR + calendar OLS)	0.1270	0.1232	+16.3	+2.63	1.135	2
block-diagonal ridge^{R}	0.1092	0.1068			0.989	1
block-diagonal ridge, without the FOMC columns	0.1081	0.1057	−1.0	−1.62	0.987	1
LightGBM	0.1090	0.1060	−0.2	−0.05	1.010	0
XGBoost	0.1092	0.1063	+0.0	+0.01	1.023	0
lasso (causally tuned)	0.1115	0.1090	+2.1	+0.69	1.043	2
lasso (fixed 10^{-4})	0.1101	0.1076	+0.8	+0.77	0.995	1
elastic net (causally tuned)	0.1134	0.1110	+3.8	+1.28	1.038	1
lasso (fixed 10^{-4}), earlier panel [`on disk, unused by the paper`]	0.1078	0.1054	−1.3	−1.11	0.997	1
per-bar linear						
per-bar ridge [`lean_vol_vix`]	0.0985	0.0966	−9.8	−1.94	0.901	7
per-bar elastic net [`all_features`]	0.1006	0.0986	−7.9	−2.10	0.925	4
per-bar lasso [`all_features`]	0.0996	0.0973	−8.8	−2.80	0.914	4
per-bar elastic net [`lean_vol_vix`], pre-tuned	0.0955	0.0937	−12.5	−3.30	0.988	3
per-bar elastic net [`all_features`], pre-tuned	0.0958	0.0942	−12.2	−2.97	0.950	4
per-bar ridge [`lean_vol_vix`], pre-tuned	0.0935	0.0916	−14.4	−3.78	0.976	3
per-bar lasso [`all_features`], pre-tuned	0.0931	0.0914	−14.7	−3.72	0.945	3
per-bar lasso [`lean_vol_vix`]	0.0958	0.0938	−12.3	−3.64	0.947	4
per-bar ridge [`all_features`]	0.1184	0.1164	+8.4	+0.72	0.907	6
per-bar elastic net [`lean_vol_vix`]	0.0968	0.0945	−11.3	−3.58	0.951	3
per-bar lasso [`lean_vol_vix`], pre-tuned	0.0964	0.0947	−11.8	−2.95	0.903	5
per-bar ridge [`all_features`], pre-tuned	0.0975	0.0958	−10.7	−2.35	0.966	4
per-bar elastic net [`baseline`], pre-tuned	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar lasso [`baseline`], pre-tuned	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar ridge [`baseline`], pre-tuned	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar OLS [`baseline`]	0.0967	0.0946	−11.5	−2.41	1.026	0
per-bar ridge [`baseline`]	0.0968	0.0946	−11.3	−2.43	1.041	0
per-bar lasso [`baseline`]	0.0956	0.0934	−12.4	−2.56	1.052	0
per-bar elastic net [`baseline`]	0.0957	0.0936	−12.4	−2.55	1.047	0
pooled twin						
pooled elastic net [`all_features`], pre-tuned	0.1040	0.1016	−4.8	−1.80	1.057	1
pooled ridge [`all_features`], pre-tuned	0.1045	0.1021	−4.3	−1.73	1.028	1
pooled elastic net [`lean_vol_vix`]	0.1029	0.1002	−5.8	−1.78	1.032	0
pooled lasso [`lean_vol_vix`]	0.1057	0.1027	−3.2	−0.95	1.066	0
pooled ridge [`lean_vol_vix`]	0.1058	0.1031	−3.2	−1.15	1.078	0
pooled elastic net [`lean_vol_vix`], pre-tuned	0.1063	0.1038	−2.6	−0.86	1.109	0
pooled ridge [`all_features`]	0.1050	0.1026	−3.8	−1.66	1.019	1
pooled lasso [`lean_vol_vix`], pre-tuned	0.1063	0.1037	−2.6	−0.85	1.110	0
pooled ridge [`lean_vol_vix`], pre-tuned	0.1066	0.1040	−2.4	−0.78	1.105	0
pooled lasso [`all_features`], pre-tuned	0.1069	0.1044	−2.1	−0.71	1.059	1
pooled elastic net [`all_features`]	0.1019	0.0995	−6.7	−2.41	1.016	1
pooled ridge [`baseline`]	0.1092	0.1060	−0.0	−0.00	1.136	0
pooled lasso [`all_features`]	0.1049	0.1025	−3.9	−1.51	1.028	1
pooled elastic net [`baseline`]	0.1087	0.1056	−0.5	−0.11	1.143	0
pooled ridge [`baseline`], pre-tuned	0.1098	0.1067	+0.6	+0.14	1.135	0
pooled elastic net [`baseline`], pre-tuned	0.1096	0.1065	+0.4	+0.09	1.156	0
pooled lasso [`baseline`]	0.1087	0.1057	−0.4	−0.11	1.139	0
pooled lasso [`baseline`], pre-tuned	0.1092	0.1062	+0.0	+0.01	1.154	0
VIX-only family						
per-bar ridge [`live_vix_only`]	0.1201	0.1179	+10.0	+0.55	0.944	5
per-bar ridge [`free_vix_only`]	0.1216	0.1194	+11.4	+0.57	0.941	6
per-bar ridge [`free_feasible`]	0.1431	0.1407	+31.0	+0.90	0.950	0
per-bar ridge [`live_vix_rvol`]	0.1156	0.1134	+5.9	+0.40	0.944	6
per-bar ridge [`free_feasible_vol`]	0.1220	0.1198	+11.7	+0.60	0.956	6
per-bar lasso [`free_vix_only`]	0.0959	0.0940	−12.2	−3.48	0.944	4
per-bar lasso [`free_feasible_vol`]	0.0949	0.0930	−13.1	−3.85	0.949	4

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
per-bar elastic net [`free_vix_only`]	0.0972	0.0950	−10.9	−3.45	0.951	3
per-bar elastic net [`live_vix_rvol`]	0.0965	0.0942	−11.6	−3.65	0.953	3
per-bar elastic net [`free_feasible_vol`]	0.0973	0.0950	−10.9	−3.60	0.946	3
per-bar lasso [`live_vix_only`]	0.0956	0.0937	−12.4	−3.55	0.947	4
per-bar lasso [`live_vix_rvol`]	0.0988	0.0961	−9.5	−3.18	0.948	4
per-bar elastic net [`live_vix_only`]	0.0975	0.0952	−10.7	−3.33	0.951	3
per-bar lasso [`free_feasible`]	0.0965	0.0945	−11.6	−3.44	0.949	0
per-bar elastic net [`free_feasible`]	0.0983	0.0959	−10.0	−3.22	0.953	0
per-bar lasso [`vix_only`]	0.0957	0.0936	−12.3	−2.66	1.054	1
per-bar ridge [`vix_only`]	0.0962	0.0939	−11.9	−2.68	1.012	3
per-bar ridge [`vix_rvol`]	0.0954	0.0930	−12.7	−2.88	0.997	2
per-bar lasso [`vix_rvol`]	0.0944	0.0922	−13.6	−2.92	1.042	1
per-bar elastic net [`vix_only`]	0.0956	0.0932	−12.5	−2.80	1.026	3
per-bar elastic net [`vix_rvol`]	0.0949	0.0925	−13.1	−2.95	1.020	2
per-bar tree (untuned)						
per-bar XGBoost [`lean_vol_vix`]	0.0943	0.0923	−13.7	−3.47	0.800	1
per-bar random forest (dev-tuned) [`all_features`]	0.0998	0.0974	−8.6	−2.27	0.855	3
per-bar LightGBM [`lean_vol_vix`]	0.0946	0.0928	−13.3	−3.15	0.825	2
per-bar random forest [`lean_vol_vix`]	0.0957	0.0935	−12.3	−3.32	0.866	1
per-bar XGBoost [`all_features`]	0.0990	0.0970	−9.3	−2.25	0.730	1
per-bar LightGBM [`all_features`]	0.0983	0.0962	−10.0	−2.36	0.825	2
per-bar random forest (dev-tuned) [`lean_vol_vix`]	0.0948	0.0924	−13.2	−3.31	0.897	2
per-bar random forest [`all_features`]	0.1006	0.0982	−7.9	−2.23	0.784	1
per-bar XGBoost [`baseline`]	0.0999	0.0974	−8.5	−1.61	1.020	0
per-bar random forest [`baseline`]	0.1006	0.0982	−7.9	−1.52	1.025	1
per-bar random forest (dev-tuned) [`baseline`]	0.1012	0.0988	−7.3	−1.32	1.036	2
per-bar LightGBM [`baseline`]	0.1018	0.0993	−6.8	−1.32	1.038	0
per-bar tree (untuned, daily refit)						
per-bar XGBoost, daily refit [`lean_vol_vix`]	0.0932	0.0911	−14.7	−3.68	0.836	1
per-bar random forest, daily refit [`lean_vol_vix`]	0.0938	0.0916	−14.1	−3.78	0.879	2
per-bar random forest (dev-tuned), daily refit [`lean_vol_vix`]	0.0947	0.0923	−13.3	−3.31	0.904	2
per-bar LightGBM, daily refit [`all_features`]	0.0961	0.0939	−12.0	−3.11	0.851	2
per-bar LightGBM, daily refit [`lean_vol_vix`]	0.0939	0.0918	−14.0	−3.49	0.869	1
per-bar XGBoost, daily refit [`all_features`]	0.0957	0.0935	−12.3	−3.06	0.827	1
per-bar random forest (dev-tuned), daily refit [`baseline`]	0.1004	0.0979	−8.1	−1.46	1.016	2
per-bar XGBoost, daily refit [`baseline`]	0.1001	0.0976	−8.4	−1.58	1.021	0
per-bar random forest (dev-tuned), daily refit [`all_features`]	0.0980	0.0954	−10.3	−2.68	0.880	3
per-bar random forest, daily refit [`all_features`]	0.0991	0.0966	−9.2	−2.59	0.827	1
per-bar LightGBM, daily refit [`baseline`]	0.1010	0.0986	−7.5	−1.44	1.030	0
per-bar random forest, daily refit [`baseline`]	0.0992	0.0968	−9.2	−1.78	1.016	1
per-bar tree (tuned)						
tuned per-bar random forest [`lean_vol_vix`], MSE-selected	0.0993	0.0971	−9.1	−2.45	0.873	1
tuned per-bar LightGBM [`all_features`], MSE-selected	0.1005	0.0982	−7.9	−1.73	0.876	0
tuned per-bar LightGBM [`lean_vol_vix`], MSE-selected	0.1044	0.1021	−4.4	−0.99	0.908	0
tuned per-bar LightGBM [`lean_vol_vix`], QLIKE-selected	0.0996	0.0977	−8.8	−2.05	0.907	0
tuned per-bar XGBoost [`all_features`], MSE-selected	0.1030	0.1007	−5.7	−1.25	0.838	0
tuned per-bar XGBoost [`all_features`], QLIKE-selected	0.1029	0.1006	−5.7	−1.27	0.837	0
tuned per-bar LightGBM [`all_features`], QLIKE-selected	0.0998	0.0975	−8.6	−1.95	0.847	0
tuned per-bar random forest [`lean_vol_vix`], QLIKE-selected	0.0970	0.0948	−11.1	−3.10	0.871	2
tuned per-bar XGBoost [`lean_vol_vix`], QLIKE-selected	0.1049	0.1027	−3.9	−0.86	0.912	0
tuned per-bar XGBoost [`lean_vol_vix`], MSE-selected	0.1041	0.1015	−4.7	−1.04	0.926	0
tuned per-bar random forest [`all_features`], QLIKE-selected	0.1029	0.1003	−5.8	−1.67	0.797	1
tuned per-bar random forest [`all_features`], MSE-selected	0.1085	0.1059	−0.6	−0.13	0.707	1
tuned per-bar XGBoost [`baseline`], MSE-selected	0.1100	0.1069	+0.7	+0.12	1.076	0
tuned per-bar LightGBM [`baseline`], MSE-selected	0.1071	0.1043	−1.9	−0.34	1.029	0
tuned per-bar LightGBM [`baseline`], QLIKE-selected	0.1027	0.1004	−5.9	−1.16	1.050	0
tuned per-bar XGBoost [`baseline`], QLIKE-selected	0.1055	0.1030	−3.4	−0.56	1.077	0
tuned per-bar random forest [`baseline`], QLIKE-selected	0.0989	0.0965	−9.4	−1.86	1.042	0
tuned per-bar random forest [`baseline`], MSE-selected	0.1089	0.1057	−0.3	−0.05	1.049	0
per-bar tree (random search, daily)						
tuned per-bar XGBoost, daily refit [`lean_vol_vix`], MSE-selected	0.1031	0.1006	−5.6	−1.27	0.927	0
tuned per-bar random forest, daily refit [`lean_vol_vix`], MSE-selected	0.0974	0.0951	−10.8	−2.88	0.885	1
tuned per-bar LightGBM, daily refit [`lean_vol_vix`], QLIKE-selected	0.0982	0.0962	−10.1	−2.37	0.912	0
tuned per-bar XGBoost, daily refit [`all_features`], MSE-selected	0.1009	0.0984	−7.6	−1.77	0.858	0

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
tuned per-bar XGBoost, daily refit [`all_features`], QLIKE-selected	0.1009	0.0985	−7.6	−1.76	0.859	0
tuned per-bar LightGBM, daily refit [`all_features`], MSE-selected	0.0990	0.0966	−9.3	−2.16	0.903	0
tuned per-bar random forest, daily refit [`lean_vol_vix`], QLIKE-selected	0.0949	0.0926	−13.1	−3.59	0.883	2
tuned per-bar XGBoost, daily refit [`lean_vol_vix`], QLIKE-selected	0.1029	0.1006	−5.8	−1.29	0.924	0
tuned per-bar LightGBM, daily refit [`all_features`], QLIKE-selected	0.0980	0.0956	−10.2	−2.47	0.880	0
tuned per-bar LightGBM, daily refit [`lean_vol_vix`], MSE-selected	0.1041	0.1015	−4.7	−1.12	0.914	0
tuned per-bar random forest, daily refit [`all_features`], MSE-selected	0.1045	0.1020	−4.3	−1.13	0.773	1
tuned per-bar random forest, daily refit [`all_features`], QLIKE-selected	0.1002	0.0976	−8.2	−2.30	0.834	1
tuned per-bar XGBoost, daily refit [`baseline`], QLIKE-selected	0.1052	0.1028	−3.6	−0.60	1.074	0
tuned per-bar LightGBM, daily refit [`baseline`], QLIKE-selected	0.1026	0.1002	−6.0	−1.19	1.039	0
tuned per-bar random forest, daily refit [`baseline`], MSE-selected	0.1090	0.1058	−0.1	−0.02	1.044	0
tuned per-bar LightGBM, daily refit [`baseline`], MSE-selected	0.1073	0.1045	−1.7	−0.30	1.028	0
tuned per-bar XGBoost, daily refit [`baseline`], MSE-selected	0.1098	0.1066	+0.6	+0.10	1.071	0
tuned per-bar random forest, daily refit [`baseline`], QLIKE-selected	0.0995	0.0971	−8.8	−1.72	1.035	0
per-bar tree (Optuna, TUNE_PER=1, best-of-50)						
Optuna per-bar random forest, TUNE_PER 1, best of 50 [`lean_vol_vix`]	0.0966	0.0941	−11.5	−3.15	0.902	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`lean_vol_vix`]	0.1020	0.0995	−6.6	−1.82	0.877	0
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`lean_vol_vix`]	0.1002	0.0978	−8.2	−2.03	0.919	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`all_features`]	0.1032	0.1010	−5.5	−1.43	0.883	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`all_features`]	0.1017	0.0993	−6.9	−1.96	0.865	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`baseline`]	0.1055	0.1028	−3.4	−0.67	1.028	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`baseline`]	0.1036	0.1011	−5.1	−0.99	0.992	0
Optuna per-bar random forest, TUNE_PER 1, best of 50 [`baseline`]	0.1015	0.0990	−7.1	−1.49	1.042	0
Optuna per-bar random forest, TUNE_PER 1, best of 50 [`all_features`]	0.1012	0.0985	−7.3	−2.01	0.868	1
per-bar tree (Optuna, TUNE_PER=5, best-of-50)						
Optuna per-bar random forest, TUNE_PER 5, best of 50 [`lean_vol_vix`]	0.0964	0.0939	−11.7	−3.25	0.887	1
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [`lean_vol_vix`]	0.0997	0.0973	−8.7	−2.22	0.887	0
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [`all_features`]	0.1005	0.0982	−7.9	−2.11	0.820	1
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [`lean_vol_vix`]	0.0999	0.0975	−8.5	−2.19	0.899	1
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [`all_features`]	0.1014	0.0994	−7.1	−1.90	0.892	0
Optuna per-bar random forest, TUNE_PER 5, best of 50 [`all_features`]	0.1012	0.0985	−7.3	−2.04	0.863	1
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [`baseline`]	0.1023	0.0999	−6.3	−1.16	0.993	0
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [`baseline`]	0.1044	0.1016	−4.4	−0.84	1.025	0
Optuna per-bar random forest, TUNE_PER 5, best of 50 [`baseline`]	0.1015	0.0990	−7.1	−1.44	1.036	0
per-bar tree (Optuna, TUNE_PER=25, best-of-50)						
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [`all_features`]	0.1064	0.1042	−2.5	−0.60	0.793	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [`all_features`]	0.1039	0.1016	−4.9	−1.09	0.749	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [`lean_vol_vix`]	0.0988	0.0968	−9.5	−2.25	0.857	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [`lean_vol_vix`]	0.0968	0.0943	−11.4	−2.77	0.962	1
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [`lean_vol_vix`]	0.0991	0.0968	−9.3	−1.98	0.921	0
Optuna per-bar random forest, TUNE_PER 25, best of 50 [`lean_vol_vix`]	0.0989	0.0964	−9.5	−2.59	0.872	1
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [`lean_vol_vix`]	0.0958	0.0934	−12.3	−3.36	0.885	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [`all_features`]	0.1036	0.1015	−5.1	−1.16	0.775	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [`all_features`]	0.1045	0.1020	−4.3	−0.98	0.849	0
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [`all_features`]	0.1023	0.0997	−6.3	−1.85	0.816	1
Optuna per-bar random forest, TUNE_PER 25, best of 50 [`all_features`]	0.1038	0.1009	−5.0	−1.46	0.819	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [`lean_vol_vix`]	0.1050	0.1026	−3.9	−0.94	0.877	1
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [`baseline`]	0.1052	0.1026	−3.7	−0.68	1.040	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [`baseline`]	0.1032	0.1007	−5.5	−1.01	1.035	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [`baseline`]	0.1034	0.1011	−5.3	−0.90	1.007	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [`baseline`]	0.1036	0.1007	−5.1	−0.98	1.046	0
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [`baseline`]	0.1008	0.0983	−7.7	−1.50	1.020	0
Optuna per-bar random forest, TUNE_PER 25, best of 50 [`baseline`]	0.1017	0.0992	−6.8	−1.38	1.031	0
per-bar tree (Optuna, TUNE_PER=250, best-of-50)						
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [`lean_vol_vix`]	0.1035	0.1010	−5.2	−1.33	0.895	0
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [`all_features`]	0.1006	0.0983	−7.8	−1.72	0.908	0
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [`lean_vol_vix`]	0.1064	0.1040	−2.5	−0.59	0.926	0
Optuna per-bar random forest, TUNE_PER 250, best of 50 [`lean_vol_vix`]	0.0953	0.0930	−12.7	−3.43	0.886	1
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [`all_features`]	0.1013	0.0991	−7.3	−1.47	0.857	0
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [`baseline`]	0.1092	0.1062	+0.0	+0.00	1.033	0
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [`baseline`]	0.1100	0.1068	+0.7	+0.12	1.039	0
Optuna per-bar random forest, TUNE_PER 250, best of 50 [`all_features`]	0.1027	0.1001	−5.9	−1.60	0.826	1
Optuna per-bar random forest, TUNE_PER 250, best of 50 [`baseline`]	0.1060	0.1036	−2.9	−0.51	1.055	0
per-bar tree (Optuna, TUNE_PER=1, best-of-25)						

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [all_features]	0.1028	0.1004	−5.8	−1.58	0.878	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [lean_vol_vix]	0.0995	0.0971	−8.9	−2.23	0.908	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [all_features]	0.1000	0.0976	−8.4	−2.33	0.865	1
Optuna per-bar random forest, TUNE_PER 1, best of 25 [lean_vol_vix]	0.0969	0.0944	−11.3	−3.16	0.895	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [baseline]	0.1037	0.1011	−5.1	−0.96	0.991	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [lean_vol_vix]	0.1002	0.0978	−8.2	−2.36	0.887	0
Optuna per-bar random forest, TUNE_PER 1, best of 25 [all_features]	0.1006	0.0979	−7.9	−2.13	0.873	1
Optuna per-bar random forest, TUNE_PER 1, best of 25 [baseline]	0.1013	0.0988	−7.3	−1.52	1.043	0
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [baseline]	0.1058	0.1029	−3.1	−0.66	1.032	0
per-bar tree (Optuna, TUNE_PER=25, best-of-25)						
Optuna per-bar random forest, TUNE_PER 25, best of 25 [lean_vol_vix]	0.0974	0.0948	−10.8	−3.02	0.867	2
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [all_features]	0.1031	0.1008	−5.6	−1.31	0.743	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [lean_vol_vix]	0.1003	0.0979	−8.2	−1.90	0.889	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [lean_vol_vix]	0.1034	0.1010	−5.3	−1.32	0.876	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [all_features]	0.1026	0.1003	−6.1	−1.43	0.807	0
Optuna per-bar random forest, TUNE_PER 25, best of 25 [all_features]	0.1022	0.0996	−6.4	−1.79	0.820	1
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [baseline]	0.1035	0.1009	−5.2	−0.93	1.018	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [baseline]	0.1040	0.1016	−4.8	−0.91	1.039	0
Optuna per-bar random forest, TUNE_PER 25, best of 25 [baseline]	0.1015	0.0990	−7.0	−1.41	1.031	0
per-bar tree (Optuna, TUNE_PER=1, best-of-10)						
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [lean_vol_vix]	0.0996	0.0974	−8.8	−2.34	0.885	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [all_features]	0.1031	0.1007	−5.6	−1.35	0.840	1
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [lean_vol_vix]	0.0998	0.0973	−8.6	−2.22	0.859	0
Optuna per-bar random forest, TUNE_PER 1, best of 10 [all_features]	0.1008	0.0981	−7.7	−2.13	0.855	1
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [all_features]	0.1031	0.1006	−5.6	−1.53	0.846	0
Optuna per-bar random forest, TUNE_PER 1, best of 10 [lean_vol_vix]	0.0975	0.0948	−10.7	−2.99	0.884	2
Optuna per-bar random forest, TUNE_PER 1, best of 10 [baseline]	0.1014	0.0989	−7.1	−1.48	1.035	0
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [baseline]	0.1051	0.1025	−3.8	−0.77	1.013	0
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [baseline]	0.1024	0.0999	−6.2	−1.24	1.002	0
per-bar tree (Optuna, TUNE_PER=25, best-of-10)						
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [lean_vol_vix]	0.1029	0.1005	−5.8	−1.54	0.780	0
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [lean_vol_vix]	0.0982	0.0959	−10.1	−2.53	0.845	1
Optuna per-bar random forest, TUNE_PER 25, best of 10 [all_features]	0.1006	0.0979	−7.9	−2.27	0.831	1
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [all_features]	0.1055	0.1031	−3.4	−0.76	0.855	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [all_features]	0.1076	0.1053	−1.4	−0.34	0.739	0
Optuna per-bar random forest, TUNE_PER 25, best of 10 [lean_vol_vix]	0.0962	0.0937	−11.9	−3.21	0.885	2
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [baseline]	0.1058	0.1032	−3.1	−0.61	1.060	0
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [baseline]	0.1030	0.1003	−5.7	−1.06	1.023	0
Optuna per-bar random forest, TUNE_PER 25, best of 10 [baseline]	0.1020	0.0995	−6.6	−1.26	1.021	0
direct rest-of-day at 15:30 (check)						
direct rest-of-day elastic net [all_features], 15:30 clock$^{=}$	0.1006	0.0986	−7.9	−2.10	0.925	4
direct rest-of-day lasso [all_features], 15:30 clock$^{=}$	0.0996	0.0973	−8.8	−2.80	0.914	4
direct rest-of-day ridge [all_features], 15:30 clock$^{=}$	0.1184	0.1164	+8.4	+0.72	0.907	6
direct rest-of-day ridge [baseline], 15:30 clock$^{=}$	0.0968	0.0946	−11.3	−2.43	1.041	0
direct rest-of-day lasso [baseline], 15:30 clock$^{=}$	0.0956	0.0934	−12.4	−2.56	1.052	0
direct rest-of-day elastic net [baseline], 15:30 clock$^{=}$	0.0957	0.0936	−12.4	−2.55	1.047	0
implied-vol representations						
per-bar elastic net [ivrep_term]	0.0966	0.0943	−11.5	−3.74	0.972	3
per-bar ridge [ivrep_term]	0.1166	0.1145	+6.8	+0.44	0.981	4
per-bar ridge [ivrep_vrp]	0.1102	0.1080	+0.9	+0.07	0.934	5
per-bar lasso [ivrep_vrp]	0.0947	0.0921	−13.2	−3.43	0.923	5
per-bar lasso [ivrep_term]	0.0954	0.0935	−12.6	−3.72	0.964	4
per-bar ridge [ivrep_all]	0.1113	0.1091	+1.9	+0.17	0.982	4
per-bar elastic net [ivrep_vrp]	0.0941	0.0919	−13.8	−4.11	0.928	3
per-bar ridge [ivrep_target_scale]	0.1124	0.1102	+2.9	+0.22	0.919	4
per-bar lasso [ivrep_innovations]	0.0983	0.0957	−10.0	−2.66	0.995	3
per-bar lasso [ivrep_target_scale]	0.0950	0.0932	−13.0	−3.17	0.870	2
per-bar ridge [ivrep_innovations]	0.1133	0.1113	+3.7	+0.27	1.007	4
per-bar lasso [ivrep_all]	0.0996	0.0970	−8.8	−2.24	0.918	3
per-bar elastic net [ivrep_all]	0.0953	0.0932	−12.7	−3.07	0.926	3
per-bar elastic net [ivrep_innovations]	0.0965	0.0944	−11.6	−2.79	1.009	3
per-bar elastic net [ivrep_target_scale]	0.0965	0.0946	−11.6	−2.94	0.875	4

Table B: scored on the forecast's own days, paired with the reference on them (550 trade days from 2022-01-03)

(continued)

Forecast	QLIKE	QLIKE, trimmed	% vs R	DM	$\overline{RV}/\overline{F}$	extreme (all sess.)
chain-period buckets						
per-bar elastic net [`ivrep_slice_slope`], 500 sessions	0.0915	0.0890	−4.1	−0.94	0.972	6
per-bar ridge [`ivrep_slice_slope`], 500 sessions	0.0895	0.0872	−6.2	−1.44	0.952	5
per-bar lasso [`ivrep_slice_slope`], 500 sessions	0.0957	0.0926	+0.3	+0.06	0.986	5
per-bar ridge [`ivslice_only`], 500 sessions	0.0944	0.0920	−1.0	−0.23	0.981	0
per-bar elastic net [`ivslice_only`], 500 sessions	0.0970	0.0947	+1.6	+0.32	0.998	0
per-bar lasso [`ivslice_only`], 500 sessions	0.0993	0.0977	+4.0	+0.53	1.003	0
per-bar lasso [`mfiv_only`], 500 sessions	0.1030	0.1011	+7.9	+1.25	1.005	0
per-bar elastic net [`mfiv_only`], 500 sessions	0.1020	0.1001	+6.8	+1.10	1.001	0
per-bar ridge [`mfiv_only`], 500 sessions	0.1017	0.0996	+6.6	+1.12	1.014	0

Table 7: The last-30-min trade: sign(s) on the straddle (nearest out-of-the-money call and put, same-day expiry, one position), entered at 15:30, held to the close; buy when the plain 16:00 forecast f^2B exceeds the 15:30 implied variance, sell otherwise. Returns per unit of premium; Sharpe annualized by $\sqrt{252}$ per trade day; $^{\times}$: the entry pays the spread (buy at the ask, sell at the bid); hit: share of days with a positive midpoint return; buys: days the rule buys. ΔSharpe: the row minus the reference R (or minus always short) on the same days, point estimates; t vs R: Newey–West HAC t statistic of the mean daily midpoint-return difference against R.

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Table A: the same 866 trade days, 2020-01-03 to 2024-04-30											
rule											
always short (no forecast)	0.20	−0.27	+0.014	−0.020	61.8	0	−1.08	−1.07	−1.71		
paper											
baseline (HAR + calendar OLS)	0.43	−0.05	+0.030	−0.004	57.6	148	−0.85	−0.85	−1.49	+0.22	+0.22
block-diagonal ridge^{R}	1.28	0.80	+0.091	+0.057	56.7	234				+1.08	+1.07
block-diagonal ridge, without the FOMC columns	1.36	0.88	+0.096	+0.062	56.9	236	+0.08	+0.08	+0.44	+1.15	+1.15
LightGBM	1.43	0.93	+0.101	+0.067	58.9	165	+0.15	+0.14	+0.27	+1.22	+1.21
XGBoost	1.26	0.77	+0.090	+0.055	60.2	162	−0.02	−0.03	−0.03	+1.06	+1.04
lasso (causally tuned)	1.50	1.01	+0.106	+0.072	59.0	184	+0.22	+0.21	+0.54	+1.30	+1.28
lasso (fixed 10^{-4})	1.06	0.58	+0.075	+0.041	56.7	246	−0.22	−0.22	−0.75	+0.85	+0.85
elastic net (causally tuned)	0.81	0.32	+0.057	+0.023	57.2	182	−0.48	−0.48	−1.09	+0.60	+0.59
lasso (fixed 10^{-4}), earlier panel [`on disk, unused by the paper`]	1.36	0.88	+0.097	+0.063	56.2	254	+0.08	+0.09	+0.42	+1.16	+1.16
per-bar linear											
per-bar ridge [`lean_vol_vix`]	2.11	1.63	+0.148	+0.115	57.7	257	+0.83	+0.83	+1.87	+1.90	+1.90
per-bar elastic net [`all_features`]	2.04	1.56	+0.144	+0.110	58.1	256	+0.76	+0.76	+1.63	+1.84	+1.84
per-bar lasso [`all_features`]	2.00	1.53	+0.141	+0.107	58.0	259	+0.72	+0.73	+1.53	+1.80	+1.80
per-bar elastic net [`lean_vol_vix`], pre-tuned	1.93	1.45	+0.136	+0.102	57.9	250	+0.65	+0.65	+1.34	+1.73	+1.72
per-bar elastic net [`all_features`], pre-tuned	1.92	1.43	+0.135	+0.102	57.7	245	+0.64	+0.63	+1.16	+1.72	+1.71
per-bar ridge [`lean_vol_vix`], pre-tuned	1.88	1.40	+0.133	+0.099	57.2	256	+0.60	+0.60	+1.22	+1.68	+1.67
per-bar lasso [`all_features`], pre-tuned	1.88	1.40	+0.132	+0.099	57.6	256	+0.60	+0.60	+1.29	+1.67	+1.67
per-bar lasso [`lean_vol_vix`]	1.73	1.25	+0.122	+0.088	57.7	247	+0.45	+0.45	+0.98	+1.53	+1.52
per-bar ridge [`all_features`]	1.71	1.22	+0.121	+0.087	57.9	240	+0.43	+0.43	+0.81	+1.50	+1.50
per-bar elastic net [`lean_vol_vix`]	1.62	1.14	+0.114	+0.081	57.0	259	+0.34	+0.34	+0.77	+1.41	+1.41
per-bar lasso [`lean_vol_vix`], pre-tuned	1.44	0.96	+0.102	+0.068	56.8	251	+0.16	+0.16	+0.29	+1.24	+1.23
per-bar ridge [`all_features`], pre-tuned	1.42	0.93	+0.100	+0.067	57.5	235	+0.14	+0.13	+0.28	+1.21	+1.21
per-bar elastic net [`baseline`], pre-tuned	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar lasso [`baseline`], pre-tuned	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar ridge [`baseline`], pre-tuned	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar OLS [`baseline`]	1.37	0.89	+0.097	+0.063	56.7	242	+0.09	+0.09	+0.16	+1.17	+1.16
per-bar ridge [`baseline`]	1.36	0.88	+0.097	+0.063	56.5	250	+0.08	+0.08	+0.15	+1.16	+1.16
per-bar lasso [`baseline`]	1.29	0.81	+0.091	+0.057	56.4	247	+0.01	+0.01	+0.01	+1.09	+1.08
per-bar elastic net [`baseline`]	1.23	0.74	+0.087	+0.053	56.2	244	−0.06	−0.06	−0.10	+1.02	+1.02
pooled twin											
pooled elastic net [`all_features`], pre-tuned	1.50	1.01	+0.106	+0.072	57.3	237	+0.22	+0.21	+0.44	+1.29	+1.29
pooled ridge [`all_features`], pre-tuned	1.44	0.96	+0.102	+0.068	57.5	235	+0.16	+0.16	+0.34	+1.24	+1.23
pooled elastic net [`lean_vol_vix`]	1.43	0.94	+0.101	+0.067	58.5	200	+0.15	+0.14	+0.36	+1.23	+1.22
pooled lasso [`lean_vol_vix`]	1.42	0.93	+0.100	+0.066	58.7	197	+0.14	+0.13	+0.32	+1.21	+1.20
pooled ridge [`lean_vol_vix`]	1.40	0.90	+0.099	+0.065	58.4	191	+0.12	+0.11	+0.26	+1.19	+1.18
pooled elastic net [`lean_vol_vix`], pre-tuned	1.29	0.80	+0.091	+0.057	57.9	196	+0.01	−0.00	+0.02	+1.09	+1.07
pooled ridge [`all_features`]	1.28	0.79	+0.091	+0.057	57.4	232	−0.00	−0.01	−0.01	+1.07	+1.07

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
pooled lasso [`lean_vol_vix`], pre-tuned	1.23	0.74	+0.087	+0.053	57.7	193	−0.05	−0.06	−0.12	+1.03	+1.01
pooled ridge [`lean_vol_vix`], pre-tuned	1.22	0.73	+0.087	+0.053	58.0	197	−0.06	−0.06	−0.14	+1.02	+1.01
pooled lasso [`all_features`], pre-tuned	1.20	0.71	+0.085	+0.051	58.0	205	−0.08	−0.09	−0.19	+1.00	+0.99
pooled elastic net [`all_features`]	1.15	0.67	+0.082	+0.048	57.5	221	−0.13	−0.13	−0.27	+0.95	+0.94
pooled ridge [`baseline`]	1.03	0.55	+0.073	+0.039	56.0	232	−0.25	−0.24	−0.50	+0.83	+0.83
pooled lasso [`all_features`]	1.00	0.52	+0.071	+0.037	56.9	224	−0.28	−0.28	−0.58	+0.80	+0.80
pooled elastic net [`baseline`]	0.95	0.47	+0.067	+0.033	55.9	227	−0.33	−0.33	−0.67	+0.74	+0.74
pooled ridge [`baseline`], pre-tuned	0.91	0.43	+0.065	+0.031	55.7	231	−0.37	−0.36	−0.76	+0.71	+0.71
pooled elastic net [`baseline`], pre-tuned	0.89	0.41	+0.063	+0.029	55.5	232	−0.39	−0.39	−0.80	+0.68	+0.68
pooled lasso [`baseline`]	0.89	0.41	+0.063	+0.029	56.1	223	−0.39	−0.39	−0.76	+0.68	+0.68
pooled lasso [`baseline`], pre-tuned	0.69	0.21	+0.049	+0.015	55.4	227	−0.59	−0.59	−1.23	+0.49	+0.49
VIX-only family											
per-bar ridge [`live_vix_only`]	2.53	2.05	+0.177	+0.144	58.8	272	+1.24	+1.25	+2.78	+2.32	+2.32
per-bar ridge [`free_vix_only`]	2.49	2.02	+0.175	+0.141	58.9	271	+1.21	+1.22	+2.65	+2.29	+2.29
per-bar ridge [`free_feasible`]	2.08	1.59	+0.146	+0.113	58.3	254	+0.79	+0.79	+1.47	+1.87	+1.86
per-bar ridge [`live_vix_rvol`]	2.04	1.56	+0.143	+0.110	57.4	272	+0.76	+0.76	+1.66	+1.83	+1.83
per-bar ridge [`free_feasible_vol`]	2.03	1.55	+0.143	+0.110	57.6	268	+0.75	+0.75	+1.51	+1.83	+1.83
per-bar lasso [`free_vix_only`]	1.91	1.43	+0.135	+0.101	58.2	251	+0.63	+0.63	+1.30	+1.71	+1.70
per-bar lasso [`free_feasible_vol`]	1.81	1.33	+0.128	+0.094	57.2	252	+0.53	+0.53	+1.11	+1.60	+1.60
per-bar elastic net [`free_vix_only`]	1.70	1.22	+0.120	+0.087	57.4	256	+0.42	+0.43	+0.98	+1.50	+1.50
per-bar elastic net [`live_vix_rvol`]	1.64	1.16	+0.116	+0.082	56.9	258	+0.36	+0.36	+0.78	+1.43	+1.43
per-bar elastic net [`free_feasible_vol`]	1.60	1.13	+0.113	+0.080	56.8	261	+0.32	+0.33	+0.76	+1.40	+1.40
per-bar lasso [`live_vix_only`]	1.58	1.09	+0.112	+0.078	58.0	245	+0.30	+0.29	+0.51	+1.38	+1.36
per-bar lasso [`live_vix_rvol`]	1.55	1.07	+0.110	+0.076	56.8	255	+0.27	+0.27	+0.61	+1.35	+1.35
per-bar elastic net [`live_vix_only`]	1.53	1.04	+0.109	+0.075	57.7	249	+0.25	+0.24	+0.46	+1.33	+1.32
per-bar lasso [`free_feasible`]	1.40	0.92	+0.099	+0.066	56.9	250	+0.12	+0.12	+0.31	+1.20	+1.20
per-bar elastic net [`free_feasible`]	1.33	0.85	+0.094	+0.060	56.0	254	+0.05	+0.05	+0.11	+1.12	+1.12
per-bar lasso [`vix_only`]	1.13	0.64	+0.080	+0.046	57.0	203	−0.15	−0.16	−0.27	+0.92	+0.92
per-bar ridge [`vix_only`]	1.09	0.61	+0.077	+0.043	56.5	214	−0.19	−0.19	−0.38	+0.89	+0.88
per-bar ridge [`vix_rvol`]	0.91	0.43	+0.065	+0.031	55.9	213	−0.37	−0.37	−0.77	+0.71	+0.71
per-bar lasso [`vix_rvol`]	0.79	0.31	+0.056	+0.022	56.5	208	−0.49	−0.49	−0.98	+0.59	+0.58
per-bar elastic net [`vix_only`]	0.79	0.31	+0.056	+0.022	55.4	229	−0.50	−0.49	−1.03	+0.58	+0.58
per-bar elastic net [`vix_rvol`]	0.74	0.26	+0.053	+0.019	55.4	231	−0.54	−0.53	−1.17	+0.54	+0.54
per-bar tree (untuned)											
per-bar XGBoost [`lean_vol_vix`]	1.78	1.30	+0.126	+0.092	57.7	253	+0.50	+0.50	+1.00	+1.58	+1.57
per-bar random forest (dev-tuned) [`all_features`]	1.75	1.27	+0.123	+0.090	56.0	288	+0.46	+0.47	+0.90	+1.54	+1.55
per-bar LightGBM [`lean_vol_vix`]	1.68	1.20	+0.119	+0.085	56.8	263	+0.40	+0.40	+0.75	+1.48	+1.48
per-bar random forest [`lean_vol_vix`]	1.60	1.11	+0.113	+0.079	56.9	264	+0.32	+0.31	+0.58	+1.40	+1.39
per-bar XGBoost [`all_features`]	1.55	1.07	+0.109	+0.076	56.7	258	+0.27	+0.27	+0.45	+1.34	+1.34
per-bar LightGBM [`all_features`]	1.37	0.88	+0.097	+0.063	56.8	247	+0.09	+0.09	+0.15	+1.16	+1.16
per-bar random forest (dev-tuned) [`lean_vol_vix`]	1.30	0.82	+0.092	+0.058	56.1	271	+0.02	+0.02	+0.04	+1.10	+1.09
per-bar random forest [`all_features`]	1.27	0.80	+0.090	+0.057	54.6	292	−0.01	+0.00	−0.02	+1.07	+1.07
per-bar XGBoost [`baseline`]	1.20	0.72	+0.085	+0.051	55.9	265	−0.08	−0.07	−0.17	+1.00	+1.00
per-bar random forest [`baseline`]	0.98	0.50	+0.069	+0.036	55.2	263	−0.30	−0.30	−0.65	+0.77	+0.77
per-bar random forest (dev-tuned) [`baseline`]	0.75	0.27	+0.053	+0.020	55.5	264	−0.53	−0.53	−0.90	+0.55	+0.55
per-bar LightGBM [`baseline`]	0.73	0.25	+0.052	+0.018	55.8	264	−0.55	−0.55	−0.95	+0.53	+0.52
per-bar tree (untuned, daily refit)											
per-bar XGBoost, daily refit [`lean_vol_vix`]	1.76	1.28	+0.124	+0.091	57.9	244	+0.48	+0.48	+0.91	+1.56	+1.55
per-bar random forest, daily refit [`lean_vol_vix`]	1.58	1.09	+0.112	+0.078	57.2	256	+0.30	+0.29	+0.54	+1.38	+1.36
per-bar random forest (dev-tuned), daily refit [`lean_vol_vix`]	1.40	0.90	+0.099	+0.065	57.3	261	+0.11	+0.10	+0.19	+1.19	+1.18
per-bar LightGBM, daily refit [`all_features`]	1.33	0.84	+0.094	+0.061	57.4	250	+0.05	+0.04	+0.08	+1.13	+1.12
per-bar LightGBM, daily refit [`lean_vol_vix`]	1.26	0.77	+0.089	+0.056	57.5	243	−0.02	−0.03	−0.03	+1.06	+1.04
per-bar XGBoost, daily refit [`all_features`]	1.12	0.63	+0.079	+0.045	56.7	250	−0.16	−0.17	−0.25	+0.91	+0.90
per-bar random forest (dev-tuned), daily refit [`baseline`]	1.07	0.59	+0.076	+0.042	56.2	258	−0.21	−0.21	−0.36	+0.87	+0.86
per-bar XGBoost, daily refit [`baseline`]	0.95	0.48	+0.068	+0.034	55.2	265	−0.33	−0.32	−0.73	+0.75	+0.75
per-bar random forest (dev-tuned), daily refit [`all_features`]	0.85	0.37	+0.060	+0.027	55.2	267	−0.43	−0.43	−0.70	+0.65	+0.64
per-bar random forest, daily refit [`all_features`]	0.85	0.37	+0.060	+0.026	54.5	277	−0.44	−0.43	−0.88	+0.64	+0.64
per-bar LightGBM, daily refit [`baseline`]	0.79	0.31	+0.056	+0.022	56.1	261	−0.49	−0.49	−0.81	+0.59	+0.59
per-bar random forest, daily refit [`baseline`]	0.66	0.19	+0.047	+0.013	55.3	264	−0.62	−0.61	−1.05	+0.46	+0.46
per-bar tree (tuned)											
tuned per-bar random forest [`lean_vol_vix`], MSE-selected	1.81	1.34	+0.128	+0.094	57.3	267	+0.53	+0.54	+1.26	+1.61	+1.61
tuned per-bar LightGBM [`all_features`], MSE-selected	1.81	1.33	+0.128	+0.094	56.9	264	+0.53	+0.54	+1.04	+1.61	+1.61
tuned per-bar LightGBM [`lean_vol_vix`], MSE-selected	1.78	1.31	+0.126	+0.092	56.0	290	+0.50	+0.51	+1.05	+1.58	+1.58
tuned per-bar LightGBM [`lean_vol_vix`], QLIKE-selected	1.72	1.25	+0.122	+0.088	57.4	262	+0.44	+0.45	+1.09	+1.52	+1.52

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
tuned per-bar XGBoost [`all_features`], MSE-selected	1.68	1.19	+0.118	+0.085	57.9	250	+0.40	+0.39	+0.66	+1.47	+1.46
tuned per-bar XGBoost [`all_features`], QLIKE-selected	1.61	1.12	+0.114	+0.080	57.7	249	+0.33	+0.32	+0.54	+1.41	+1.40
tuned per-bar LightGBM [`all_features`], QLIKE-selected	1.54	1.06	+0.109	+0.075	56.6	269	+0.26	+0.26	+0.48	+1.33	+1.33
tuned per-bar random forest [`lean_vol_vix`], QLIKE-selected	1.49	1.00	+0.105	+0.071	56.7	262	+0.20	+0.20	+0.37	+1.28	+1.27
tuned per-bar XGBoost [`lean_vol_vix`], QLIKE-selected	1.35	0.88	+0.096	+0.062	55.2	269	+0.07	+0.08	+0.14	+1.15	+1.15
tuned per-bar XGBoost [`lean_vol_vix`], MSE-selected	1.24	0.76	+0.088	+0.054	55.0	255	−0.04	−0.04	−0.06	+1.04	+1.04
tuned per-bar random forest [`all_features`], QLIKE-selected	1.24	0.76	+0.088	+0.054	55.2	273	−0.04	−0.03	−0.08	+1.03	+1.04
tuned per-bar random forest [`all_features`], MSE-selected	1.12	0.64	+0.079	+0.046	55.5	262	−0.16	−0.16	−0.30	+0.91	+0.92
tuned per-bar XGBoost [`baseline`], MSE-selected	0.99	0.51	+0.070	+0.036	53.7	290	−0.29	−0.29	−0.62	+0.78	+0.79
tuned per-bar LightGBM [`baseline`], MSE-selected	0.95	0.48	+0.068	+0.034	54.8	278	−0.33	−0.32	−0.66	+0.75	+0.75
tuned per-bar LightGBM [`baseline`], QLIKE-selected	0.94	0.46	+0.067	+0.033	54.5	279	−0.34	−0.34	−0.71	+0.74	+0.74
tuned per-bar XGBoost [`baseline`], QLIKE-selected	0.92	0.44	+0.066	+0.032	55.7	265	−0.36	−0.36	−0.60	+0.72	+0.72
tuned per-bar random forest [`baseline`], QLIKE-selected	0.78	0.30	+0.055	+0.021	55.7	269	−0.50	−0.50	−0.84	+0.57	+0.57
tuned per-bar random forest [`baseline`], MSE-selected	0.65	0.17	+0.046	+0.012	54.7	285	−0.63	−0.63	−1.06	+0.44	+0.44
per-bar tree (random search, daily)											
tuned per-bar XGBoost, daily refit [`lean_vol_vix`], MSE-selected	1.95	1.47	+0.137	+0.104	57.5	257	+0.67	+0.67	+1.37	+1.74	+1.75
tuned per-bar random forest, daily refit [`lean_vol_vix`], MSE-selected	1.86	1.38	+0.131	+0.097	57.2	268	+0.57	+0.58	+1.37	+1.65	+1.65
tuned per-bar LightGBM, daily refit [`lean_vol_vix`], QLIKE-selected	1.78	1.30	+0.126	+0.092	57.6	250	+0.50	+0.50	+1.23	+1.58	+1.58
tuned per-bar XGBoost, daily refit [`all_features`], MSE-selected	1.77	1.27	+0.125	+0.091	58.7	241	+0.48	+0.48	+0.82	+1.56	+1.55
tuned per-bar XGBoost, daily refit [`all_features`], QLIKE-selected	1.76	1.27	+0.124	+0.091	58.7	241	+0.48	+0.47	+0.82	+1.56	+1.55
tuned per-bar LightGBM, daily refit [`all_features`], MSE-selected	1.61	1.12	+0.114	+0.080	57.7	249	+0.33	+0.32	+0.59	+1.40	+1.40
tuned per-bar random forest, daily refit [`lean_vol_vix`], QLIKE-selected	1.47	0.98	+0.104	+0.070	57.3	261	+0.19	+0.18	+0.34	+1.26	+1.26
tuned per-bar XGBoost, daily refit [`lean_vol_vix`], QLIKE-selected	1.28	0.80	+0.091	+0.057	56.2	260	−0.00	−0.00	−0.00	+1.07	+1.07
tuned per-bar LightGBM, daily refit [`all_features`], QLIKE-selected	1.28	0.79	+0.090	+0.057	57.5	255	−0.00	−0.01	−0.01	+1.07	+1.06
tuned per-bar LightGBM, daily refit [`lean_vol_vix`], MSE-selected	1.25	0.77	+0.089	+0.055	55.7	277	−0.03	−0.03	−0.05	+1.05	+1.05
tuned per-bar random forest, daily refit [`all_features`], MSE-selected	1.18	0.70	+0.083	+0.050	56.5	256	−0.10	−0.10	−0.20	+0.97	+0.97
tuned per-bar random forest, daily refit [`all_features`], QLIKE-selected	1.13	0.64	+0.080	+0.046	56.4	253	−0.15	−0.15	−0.30	+0.92	+0.92
tuned per-bar XGBoost, daily refit [`baseline`], QLIKE-selected	1.09	0.61	+0.077	+0.043	55.4	269	−0.20	−0.19	−0.38	+0.88	+0.88
tuned per-bar LightGBM, daily refit [`baseline`], QLIKE-selected	0.90	0.43	+0.064	+0.030	55.1	276	−0.38	−0.37	−0.79	+0.70	+0.70
tuned per-bar random forest, daily refit [`baseline`], MSE-selected	0.68	0.20	+0.048	+0.014	54.5	293	−0.61	−0.60	−0.95	+0.47	+0.47
tuned per-bar LightGBM, daily refit [`baseline`], MSE-selected	0.64	0.16	+0.045	+0.012	55.1	278	−0.64	−0.64	−1.08	+0.44	+0.44
tuned per-bar XGBoost, daily refit [`baseline`], MSE-selected	0.57	0.09	+0.041	+0.007	53.6	287	−0.71	−0.70	−1.48	+0.37	+0.37
tuned per-bar random forest, daily refit [`baseline`], QLIKE-selected	0.46	−0.02	+0.033	−0.001	54.5	273	−0.82	−0.82	−1.31	+0.25	+0.26
per-bar tree (Optuna, TUNE_PER=1, best-of-50)											
Optuna per-bar random forest, TUNE_PER 1, best of 50 [`lean_vol_vix`]	1.50	1.01	+0.106	+0.072	57.5	257	+0.22	+0.21	+0.41	+1.30	+1.29
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`lean_vol_vix`]	1.37	0.88	+0.097	+0.063	57.4	250	+0.08	+0.08	+0.14	+1.16	+1.15
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`lean_vol_vix`]	1.28	0.79	+0.091	+0.057	57.5	255	−0.00	−0.01	−0.00	+1.08	+1.07
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`all_features`]	1.23	0.75	+0.087	+0.054	58.1	246	−0.05	−0.05	−0.07	+1.03	+1.02
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`all_features`]	1.21	0.72	+0.085	+0.052	57.0	251	−0.08	−0.08	−0.12	+1.00	+0.99
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [`baseline`]	0.95	0.47	+0.067	+0.033	55.8	274	−0.33	−0.33	−0.54	+0.74	+0.74
Optuna per-bar XGBoost, TUNE_PER 1, best of 50 [`baseline`]	0.80	0.32	+0.057	+0.023	55.2	265	−0.48	−0.48	−0.81	+0.60	+0.60

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Optuna per-bar random forest, TUNE_PER 1, best of 50 [baseline]	0.64	0.16	+0.045	+0.012	54.3	267	−0.64	−0.63	−1.33	+0.44	+0.44
Optuna per-bar random forest, TUNE_PER 1, best of 50 [all_features]	0.59	0.12	+0.042	+0.008	55.3	258	−0.69	−0.68	−1.01	+0.39	+0.39
per-bar tree (Optuna, TUNE_PER=5, best-of-50)											
Optuna per-bar random forest, TUNE_PER 5, best of 50 [lean_vol_vix]	1.48	0.99	+0.104	+0.071	57.5	257	+0.19	+0.19	+0.35	+1.27	+1.26
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [lean_vol_vix]	1.28	0.79	+0.091	+0.057	57.4	250	+0.00	−0.00	+0.01	+1.08	+1.07
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [all_features]	1.16	0.67	+0.082	+0.048	56.5	260	−0.13	−0.13	−0.21	+0.95	+0.95
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [lean_vol_vix]	0.99	0.51	+0.070	+0.037	55.3	250	−0.29	−0.29	−0.50	+0.79	+0.78
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [all_features]	0.96	0.48	+0.068	+0.034	57.4	240	−0.32	−0.32	−0.51	+0.76	+0.75
Optuna per-bar random forest, TUNE_PER 5, best of 50 [all_features]	0.86	0.38	+0.061	+0.027	55.9	257	−0.42	−0.42	−0.65	+0.66	+0.65
Optuna per-bar XGBoost, TUNE_PER 5, best of 50 [baseline]	0.80	0.32	+0.057	+0.023	55.3	272	−0.48	−0.48	−0.82	+0.60	+0.60
Optuna per-bar LightGBM, TUNE_PER 5, best of 50 [baseline]	0.47	−0.01	+0.033	−0.001	54.7	271	−0.82	−0.81	−1.41	+0.26	+0.26
Optuna per-bar random forest, TUNE_PER 5, best of 50 [baseline]	0.41	−0.07	+0.029	−0.005	54.4	266	−0.87	−0.87	−1.44	+0.20	+0.21
per-bar tree (Optuna, TUNE_PER=25, best-of-50)											
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [all_features]	1.78	1.30	+0.126	+0.092	57.7	257	+0.50	+0.50	+0.89	+1.58	+1.57
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	1.52	1.03	+0.108	+0.074	56.8	259	+0.24	+0.23	+0.41	+1.32	+1.31
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	1.49	1.00	+0.105	+0.072	58.1	266	+0.21	+0.20	+0.33	+1.29	+1.27
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	1.27	0.79	+0.090	+0.056	56.1	243	−0.01	−0.01	−0.02	+1.07	+1.06
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [lean_vol_vix]	1.27	0.78	+0.090	+0.056	57.6	262	−0.01	−0.02	−0.01	+1.07	+1.06
Optuna per-bar random forest, TUNE_PER 25, best of 50 [lean_vol_vix]	1.26	0.78	+0.089	+0.056	56.5	268	−0.02	−0.02	−0.04	+1.06	+1.05
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [lean_vol_vix]	1.24	0.75	+0.088	+0.054	56.6	259	−0.04	−0.05	−0.08	+1.03	+1.03
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [all_features]	1.23	0.74	+0.087	+0.053	56.0	262	−0.06	−0.06	−0.09	+1.02	+1.01
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	1.16	0.68	+0.082	+0.048	56.4	261	−0.12	−0.12	−0.25	+0.95	+0.95
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [all_features]	1.07	0.59	+0.076	+0.042	55.9	267	−0.21	−0.20	−0.37	+0.87	+0.87
Optuna per-bar random forest, TUNE_PER 25, best of 50 [all_features]	1.02	0.55	+0.073	+0.039	55.5	264	−0.26	−0.25	−0.44	+0.82	+0.82
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [lean_vol_vix]	0.85	0.37	+0.060	+0.027	55.3	264	−0.43	−0.43	−0.68	+0.65	+0.65
Optuna per-bar LightGBM, TUNE_PER 25, best of 50 [baseline]	0.82	0.34	+0.058	+0.025	55.5	266	−0.46	−0.46	−0.82	+0.62	+0.62
Optuna per-bar XGBoost, TUNE_PER 25, best of 50 [baseline]	0.68	0.20	+0.048	+0.014	54.8	270	−0.61	−0.60	−1.02	+0.47	+0.47
Optuna per-bar XGBoost, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.65	0.17	+0.046	+0.012	54.8	280	−0.64	−0.63	−1.07	+0.44	+0.44
Optuna per-bar LightGBM, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.55	0.07	+0.039	+0.005	55.8	256	−0.73	−0.73	−1.27	+0.35	+0.35
Optuna per-bar random forest, TUNE_PER 25, best of 50, QLIKE-selected [baseline]	0.52	0.04	+0.037	+0.003	55.0	263	−0.77	−0.76	−1.31	+0.31	+0.31
Optuna per-bar random forest, TUNE_PER 25, best of 50 [baseline]	0.45	−0.03	+0.032	−0.002	54.6	270	−0.83	−0.83	−1.45	+0.24	+0.25
per-bar tree (Optuna, TUNE_PER=250, best-of-50)											
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [lean_vol_vix]	1.77	1.29	+0.125	+0.092	56.4	253	+0.49	+0.50	+1.01	+1.57	+1.57

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [all_features]	1.55	1.08	+0.110	+0.076	56.4	257	+0.27	+0.28	+0.61	+1.35	+1.35
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [lean_vol_vix]	1.52	1.03	+0.107	+0.074	57.3	265	+0.24	+0.23	+0.40	+1.31	+1.31
Optuna per-bar random forest, TUNE_PER 250, best of 50 [lean_vol_vix]	1.29	0.80	+0.091	+0.058	57.4	252	+0.01	+0.00	+0.02	+1.09	+1.07
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [all_features]	1.24	0.76	+0.088	+0.054	55.5	252	−0.04	−0.04	−0.09	+1.04	+1.04
Optuna per-bar XGBoost, TUNE_PER 250, best of 50 [baseline]	1.22	0.74	+0.087	+0.053	56.0	276	−0.06	−0.06	−0.12	+1.02	+1.02
Optuna per-bar LightGBM, TUNE_PER 250, best of 50 [baseline]	1.08	0.61	+0.077	+0.043	54.2	284	−0.20	−0.19	−0.41	+0.88	+0.88
Optuna per-bar random forest, TUNE_PER 250, best of 50 [all_features]	0.94	0.47	+0.067	+0.033	55.3	270	−0.34	−0.33	−0.57	+0.74	+0.74
Optuna per-bar random forest, TUNE_PER 250, best of 50 [baseline]	0.64	0.17	+0.046	+0.012	55.0	273	−0.64	−0.63	−1.09	+0.44	+0.44
per-bar tree (Optuna, TUNE_PER=1, best-of-25)											
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [all_features]	1.26	0.77	+0.089	+0.055	57.6	250	−0.02	−0.03	−0.04	+1.05	+1.04
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [lean_vol_vix]	1.25	0.77	+0.089	+0.055	57.5	261	−0.03	−0.03	−0.05	+1.05	+1.04
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [all_features]	1.13	0.65	+0.080	+0.046	57.5	253	−0.15	−0.15	−0.27	+0.93	+0.92
Optuna per-bar random forest, TUNE_PER 1, best of 25 [lean_vol_vix]	1.06	0.57	+0.075	+0.041	56.8	253	−0.22	−0.23	−0.39	+0.85	+0.85
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [baseline]	0.86	0.38	+0.061	+0.027	55.3	270	−0.42	−0.42	−0.71	+0.66	+0.66
Optuna per-bar XGBoost, TUNE_PER 1, best of 25 [lean_vol_vix]	0.81	0.33	+0.058	+0.024	55.4	255	−0.47	−0.47	−0.76	+0.61	+0.60
Optuna per-bar random forest, TUNE_PER 1, best of 25 [all_features]	0.57	0.09	+0.040	+0.007	55.2	255	−0.71	−0.71	−1.04	+0.37	+0.37
Optuna per-bar random forest, TUNE_PER 1, best of 25 [baseline]	0.54	0.06	+0.038	+0.004	53.8	273	−0.74	−0.74	−1.51	+0.33	+0.34
Optuna per-bar LightGBM, TUNE_PER 1, best of 25 [baseline]	0.46	−0.02	+0.033	−0.001	53.6	275	−0.82	−0.82	−1.42	+0.26	+0.26
per-bar tree (Optuna, TUNE_PER=25, best-of-25)											
Optuna per-bar random forest, TUNE_PER 25, best of 25 [lean_vol_vix]	1.31	0.82	+0.093	+0.059	56.7	266	+0.03	+0.02	+0.05	+1.10	+1.10
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [all_features]	1.22	0.74	+0.087	+0.053	56.4	259	−0.06	−0.06	−0.09	+1.02	+1.01
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [lean_vol_vix]	1.10	0.61	+0.078	+0.044	56.8	263	−0.18	−0.19	−0.33	+0.89	+0.89
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [lean_vol_vix]	1.09	0.61	+0.077	+0.044	56.6	259	−0.19	−0.19	−0.31	+0.89	+0.88
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [all_features]	1.08	0.59	+0.076	+0.043	56.8	243	−0.20	−0.21	−0.32	+0.88	+0.87
Optuna per-bar random forest, TUNE_PER 25, best of 25 [all_features]	0.80	0.32	+0.057	+0.023	55.8	260	−0.48	−0.48	−0.74	+0.60	+0.59
Optuna per-bar XGBoost, TUNE_PER 25, best of 25 [baseline]	0.73	0.25	+0.052	+0.018	55.3	264	−0.55	−0.55	−0.94	+0.53	+0.52
Optuna per-bar LightGBM, TUNE_PER 25, best of 25 [baseline]	0.53	0.05	+0.037	+0.004	54.8	268	−0.76	−0.75	−1.37	+0.32	+0.32
Optuna per-bar random forest, TUNE_PER 25, best of 25 [baseline]	0.33	−0.15	+0.023	−0.010	54.0	275	−0.95	−0.94	−1.69	+0.13	+0.13
per-bar tree (Optuna, TUNE_PER=1, best-of-10)											
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [lean_vol_vix]	1.57	1.08	+0.111	+0.077	57.2	254	+0.29	+0.29	+0.60	+1.36	+1.36
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [all_features]	1.51	1.02	+0.107	+0.073	57.5	265	+0.22	+0.22	+0.37	+1.30	+1.29
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [lean_vol_vix]	1.46	0.97	+0.103	+0.070	56.8	249	+0.18	+0.17	+0.30	+1.26	+1.24
Optuna per-bar random forest, TUNE_PER 1, best of 10 [all_features]	1.40	0.92	+0.099	+0.065	56.1	261	+0.12	+0.12	+0.20	+1.19	+1.19

(continued)

Forecast	Sharpe	Sharpe$^{\times}$	mean	mean$^{\times}$	hit %	buys	ΔSharpe vs R	ΔSharpe$^{\times}$ vs R	t vs R	ΔSharpe vs short	ΔSharpe$^{\times}$ vs short
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [all_features]	1.29	0.80	+0.091	+0.058	56.8	277	+0.01	+0.00	+0.01	+1.09	+1.08
Optuna per-bar random forest, TUNE_PER 1, best of 10 [lean_vol_vix]	1.10	0.61	+0.078	+0.044	56.4	263	−0.18	−0.19	−0.33	+0.89	+0.89
Optuna per-bar random forest, TUNE_PER 1, best of 10 [baseline]	0.79	0.31	+0.056	+0.022	54.5	275	−0.49	−0.48	−1.03	+0.59	+0.59
Optuna per-bar LightGBM, TUNE_PER 1, best of 10 [baseline]	0.73	0.25	+0.052	+0.018	54.4	278	−0.55	−0.54	−0.91	+0.53	+0.53
Optuna per-bar XGBoost, TUNE_PER 1, best of 10 [baseline]	0.59	0.11	+0.042	+0.008	54.4	274	−0.69	−0.69	−1.17	+0.39	+0.39
per-bar tree (Optuna, TUNE_PER=25, best-of-10)											
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [lean_vol_vix]	1.71	1.23	+0.121	+0.087	56.9	258	+0.43	+0.43	+1.00	+1.51	+1.51
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [lean_vol_vix]	1.67	1.18	+0.118	+0.084	56.9	268	+0.38	+0.38	+0.79	+1.46	+1.46
Optuna per-bar random forest, TUNE_PER 25, best of 10 [all_features]	1.58	1.11	+0.112	+0.078	56.8	263	+0.30	+0.31	+0.61	+1.38	+1.38
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [all_features]	1.50	1.02	+0.106	+0.073	56.6	273	+0.22	+0.22	+0.42	+1.30	+1.30
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [all_features]	1.48	0.99	+0.105	+0.071	55.9	269	+0.20	+0.20	+0.34	+1.27	+1.27
Optuna per-bar random forest, TUNE_PER 25, best of 10 [lean_vol_vix]	1.41	0.93	+0.100	+0.066	57.2	256	+0.13	+0.13	+0.24	+1.21	+1.20
Optuna per-bar LightGBM, TUNE_PER 25, best of 10 [baseline]	0.70	0.22	+0.050	+0.016	55.3	282	−0.58	−0.58	−1.00	+0.50	+0.50
Optuna per-bar XGBoost, TUNE_PER 25, best of 10 [baseline]	0.48	0.01	+0.034	+0.000	54.3	279	−0.80	−0.79	−1.35	+0.28	+0.28
Optuna per-bar random forest, TUNE_PER 25, best of 10 [baseline]	0.29	−0.19	+0.020	−0.014	54.0	267	−1.00	−0.99	−1.63	+0.08	+0.09
direct rest-of-day at 15:30 (check)											
direct rest-of-day elastic net [all_features], 15:30 clock$^{=}$	2.04	1.56	+0.144	+0.110	58.1	256	+0.76	+0.76	+1.63	+1.84	+1.84
direct rest-of-day lasso [all_features], 15:30 clock$^{=}$	2.00	1.53	+0.141	+0.107	58.0	259	+0.72	+0.73	+1.53	+1.80	+1.80
direct rest-of-day ridge [all_features], 15:30 clock$^{=}$	1.71	1.22	+0.121	+0.087	57.9	240	+0.43	+0.43	+0.81	+1.50	+1.50
direct rest-of-day ridge [baseline], 15:30 clock$^{=}$	1.36	0.88	+0.097	+0.063	56.5	250	+0.08	+0.08	+0.15	+1.16	+1.16
direct rest-of-day lasso [baseline], 15:30 clock$^{=}$	1.29	0.81	+0.091	+0.057	56.4	247	+0.01	+0.01	+0.01	+1.09	+1.08
direct rest-of-day elastic net [baseline], 15:30 clock$^{=}$	1.23	0.74	+0.087	+0.053	56.2	244	−0.06	−0.06	−0.10	+1.02	+1.02
implied-vol representations											
per-bar elastic net [ivrep_term]	1.90	1.42	+0.134	+0.100	57.5	259	+0.62	+0.62	+1.35	+1.70	+1.69
per-bar ridge [ivrep_term]	1.87	1.38	+0.132	+0.098	58.1	262	+0.58	+0.58	+1.23	+1.66	+1.65
per-bar ridge [ivrep_vrp]	1.83	1.35	+0.129	+0.095	57.5	269	+0.55	+0.55	+1.01	+1.62	+1.62
per-bar lasso [ivrep_vrp]	1.66	1.18	+0.117	+0.084	56.5	268	+0.38	+0.38	+0.82	+1.45	+1.46
per-bar lasso [ivrep_term]	1.64	1.16	+0.116	+0.082	57.3	247	+0.36	+0.36	+0.77	+1.44	+1.43
per-bar ridge [ivrep_all]	1.61	1.13	+0.114	+0.080	56.7	264	+0.33	+0.33	+0.62	+1.41	+1.40
per-bar elastic net [ivrep_vrp]	1.59	1.11	+0.113	+0.079	56.8	259	+0.31	+0.31	+0.70	+1.39	+1.39
per-bar ridge [ivrep_target_scale]	1.58	1.10	+0.112	+0.078	56.5	256	+0.30	+0.30	+0.56	+1.38	+1.38
per-bar lasso [ivrep_innovations]	1.54	1.06	+0.109	+0.075	56.5	252	+0.26	+0.26	+0.52	+1.33	+1.33
per-bar lasso [ivrep_target_scale]	1.53	1.05	+0.108	+0.075	55.9	253	+0.25	+0.25	+0.50	+1.33	+1.33
per-bar ridge [ivrep_innovations]	1.50	1.02	+0.106	+0.072	56.8	267	+0.22	+0.22	+0.44	+1.30	+1.29
per-bar lasso [ivrep_all]	1.49	1.02	+0.106	+0.072	56.0	260	+0.21	+0.22	+0.45	+1.29	+1.29
per-bar elastic net [ivrep_all]	1.43	0.95	+0.101	+0.067	56.2	250	+0.15	+0.15	+0.29	+1.23	+1.22
per-bar elastic net [ivrep_innovations]	1.40	0.91	+0.099	+0.065	56.9	242	+0.12	+0.11	+0.22	+1.19	+1.19
per-bar elastic net [ivrep_target_scale]	1.25	0.77	+0.088	+0.055	55.9	249	−0.03	−0.03	−0.07	+1.05	+1.04
Table B: scored on the forecast's own days, paired with the reference on them (550 trade days from 2022-01-03)											
chain-period buckets											
per-bar elastic net [ivrep_slice_slope], 500 sessions	2.13	1.75	+0.142	+0.118	59.5	148	+0.52	+0.51	+0.92	+1.49	+1.48
per-bar ridge [ivrep_slice_slope], 500 sessions	2.09	1.71	+0.139	+0.115	58.9	165	+0.48	+0.48	+0.89	+1.45	+1.45
per-bar lasso [ivrep_slice_slope], 500 sessions	1.90	1.53	+0.127	+0.103	59.8	154	+0.30	+0.29	+0.52	+1.26	+1.26
per-bar ridge [ivslice_only], 500 sessions	1.35	0.98	+0.091	+0.067	58.5	143	−0.25	−0.26	−0.43	+0.71	+0.71
per-bar elastic net [ivslice_only], 500 sessions	1.33	0.96	+0.089	+0.065	58.7	140	−0.28	−0.28	−0.43	+0.69	+0.69
per-bar lasso [ivslice_only], 500 sessions	1.08	0.71	+0.072	+0.048	57.8	141	−0.53	−0.53	−0.70	+0.44	+0.44
per-bar lasso [mfiv_only], 500 sessions	1.00	0.63	+0.067	+0.043	57.6	148	−0.61	−0.61	−0.79	+0.35	+0.36
per-bar elastic net [mfiv_only], 500 sessions	0.97	0.60	+0.065	+0.041	57.1	147	−0.64	−0.63	−0.85	+0.33	+0.34
per-bar ridge [mfiv_only], 500 sessions	0.71	0.35	+0.048	+0.024	57.1	155	−0.90	−0.89	−1.33	+0.07	+0.08

Table 8: VRP sizing on the deck's scorer, 865 days from the second deck day, midpoint fill. The size on day t uses every earlier deck day back to the first: robust clip $q_t = \mathrm{clip}(s_t/\tilde{m}_t, -1, 1)$ with $\tilde{m}_t$ the median of $|s|$ on the earlier days, and the rank of $|s_t|$, sign kept. Δ is the rule's Sharpe minus sign(s) on the same days (point estimate). These Sharpes are the deck's own scorer, not the plain-forecast Sharpes of the trade table.

forecast	sign(s)	robust clip		rank	
	Sharpe	Sharpe	Δ vs sign(s)	Sharpe	Δ vs sign(s)
baseline (HAR + calendar OLS)	0.44	0.83	+0.38	0.91	+0.47
block-diagonal ridge^{R}	1.30	1.22	−0.08	1.06	−0.24
block-diagonal ridge, without the FOMC columns	1.37	1.25	−0.12	1.05	−0.33
LightGBM	1.41	1.68	+0.26	1.48	+0.07
XGBoost	1.25	1.75	+0.50	1.54	+0.29
lasso (causally tuned)	1.52	1.18	−0.34	1.05	−0.47
lasso (fixed 10^{-4})	1.08	1.19	+0.11	1.04	−0.04
elastic net (causally tuned)	0.82	0.91	+0.09	0.83	+0.01
per-bar ridge [`baseline`]	1.38	1.19	−0.19	1.17	−0.21
per-bar ridge [`all_features`]	1.73	1.57	−0.15	1.39	−0.33
per-bar lasso [`all_features`]	2.02	1.67	−0.35	1.53	−0.49
per-bar elastic net [`all_features`]	2.06	1.61	−0.45	1.61	−0.45

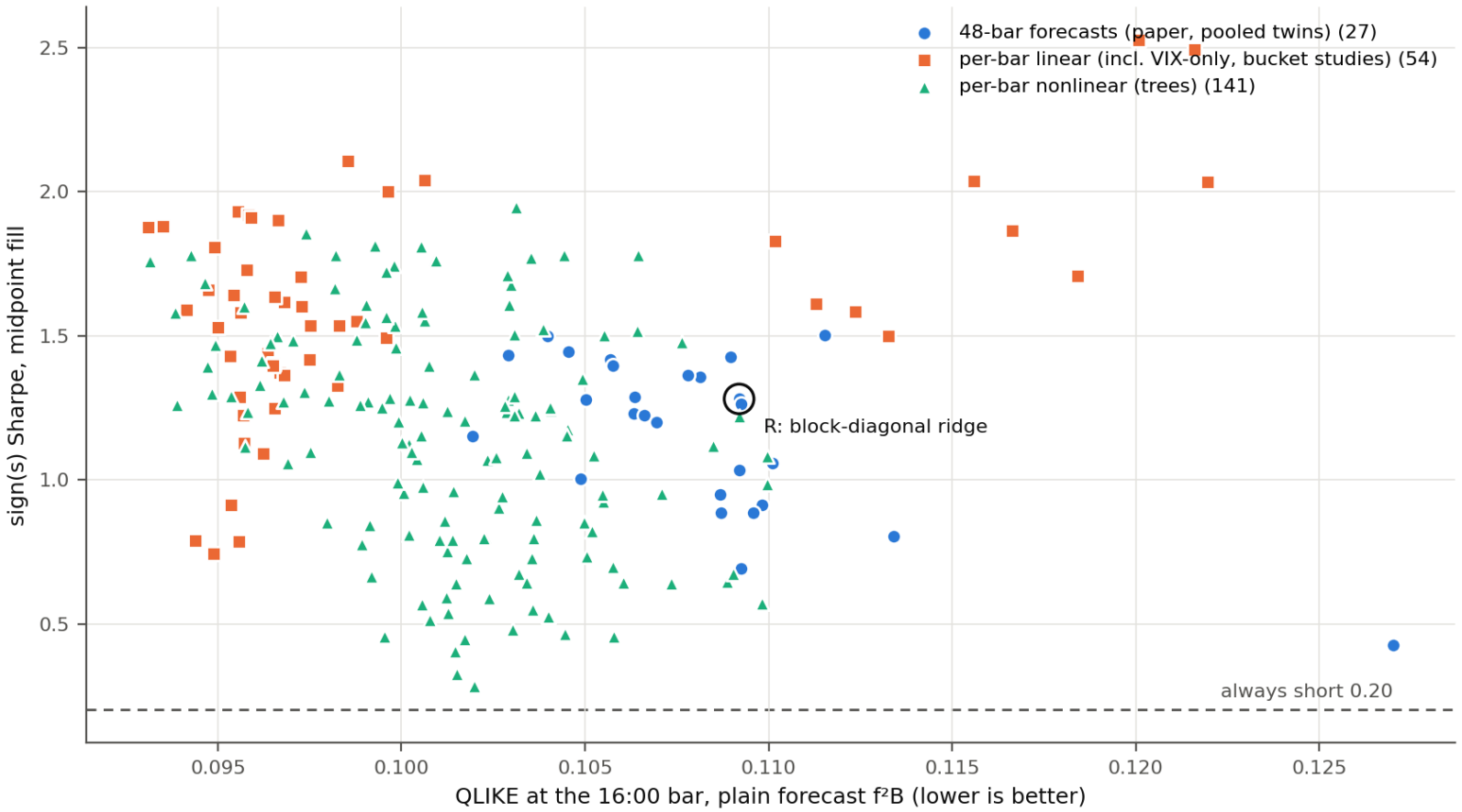


Figure 1: 16:00 QLIKE of the plain forecast f^2B against the sign(s) Sharpe ratio (midpoint fill), table A (check rows and identical forecasts left out); the dashed line is always short. R = reference, H = recommended headline. 1 forecasts lie beyond the plotted QLIKE range (upper quartile + 3 interquartile ranges = 0.127) and are not drawn: per-bar ridge [free_feasible] (QLIKE 0.143, Sharpe 2.08).